Um breve tutorial sobre dplyr com ggplot2

true

10/03/2023

Sumário

1 Introdução **5**

1.1 Alguns atalhos no Rstudio 5

1.2 Carrega pacotes a serem usados 5

1.3 Descrição dos dados `mpg` 6

1.4 Propaganda 1 (Gustavo Jun Yakushiji) 7

1.5 Propaganda 2 (Gustavo Jun Yakushiji) 8

1.6 Propaganda 3 8

2 dplyr (60 minutos) **9**

2.1 Lista de funções do pacote dplyr 9

2.2 Operador Pipe 11

2.3 select() para colunas 12

2.4 rename() 16

2.5 mutate() para colunas 17

2.6 transmute() 18

2.7 filter() para linhas 19

2.8 slice() para linhas 23

2.9 arrange() para linhas 24

2.10 distinct() para linhas 27

2.11 summarise() 29

2.12 group_by() 30

2.13 count() 31

2.14 sample_n() 33

2.15 sample_frac() 34

3 ggplot2 (60 minutos) 35

3.1 Lista de funções do pacote ggplot2 35
3.2 Primeiros passos usando geom_point 41
3.3 smooth, boxplot, histogram 58
3.4 gridExtra e patchwork 62
3.5 bar, col, density, density2d 69
3.6 facet_grid, facet_grid 76
3.7 stat_function 83
3.8 stat_summary 83
3.9 theme_*() 84
3.10 Gráfico de perfis (Spaguetti plot) 85
3.11 plotly 87
3.12 esquisse 87
3.13 Exemplo esquisse 88

4 Referências (on-line) 89

4.1 Livros 89
4.2 Tidyverse: links do dplyr, ggplot2 e magrittr 89
4.3 cheat sheet 89

Capítulo 1

Introdução

Palestra do *XIV Programa de Verão DES-ICET/UFLA - 2023*

1.1 Alguns atalhos no Rstudio

Para considerar

> Operador Pipe (%>%): Ctrl + Shift + M (Windows) ou Cmd + Shift + M (Mac).
>
> Criar novos chunks: Ctrl + Alt + I (Windows) ou Cmd + Option + I (Mac).

1.2 Carrega pacotes a serem usados

```
#install.packages("tidyverse")
#install.packages("dplyr")
#install.packages("tidyr")
#install.packages("ggplot2")

# Manipulação de dados
library(dplyr)

# Visualização de gráficos
library(ggplot2)
```

```
library(gridExtra)
library(patchwork)
library(plotly)
library(esquisse)

# Para dados gráfico de perfis
library(nlme)
```

1.3 Descrição dos dados `mpg`

Dados de economia de combustível de 1999 a 2008 para *38 modelos populares de carros*. Este conjunto de dados contém um subconjunto dos dados de economia de combustível que a EPA disponibiliza em *https://fueleconomy.gov/*. Ele contém apenas modelos que tiveram um novo lançamento a cada ano entre 1999 e 2008 - isso foi usado como um substituto para a popularidade do carro. Um *data frame* com 234 linhas e 11 variáveis:

- *manufacturer* nome do fabricante
- *model* nome do modelo
- *displ* cilindrada do motor, em litros
- *year* ano de fabricação
- *cyl* número de cilindros
- *trans* tipo de transmissão
- *drv* o tipo de trem de força, onde **f = tração dianteira**, **r = tração traseira** e **4 = 4wd**
- *cty* milhas urbanas por galão
- *hwy* milhas rodoviárias por galão
- *fl* tipo de combustível
- *class* "tipo" de carro

```
#help("mpg")
dados <- mpg
glimpse(dados)
```

```
## Rows: 234
## Columns: 11
## $ manufacturer <chr> "audi", "audi", "audi", "audi", "audi", "audi", "audi", "~
## $ model        <chr> "a4", "a4", "a4", "a4", "a4", "a4", "a4", "a4 quattro", "~
## $ displ        <dbl> 1.8, 1.8, 2.0, 2.0, 2.8, 2.8, 3.1, 1.8, 1.8, 2.0, 2.0, 2.~
## $ year         <int> 1999, 1999, 2008, 2008, 1999, 1999, 2008, 1999, 1999, 200~
## $ cyl          <int> 4, 4, 4, 4, 6, 6, 6, 4, 4, 4, 4, 6, 6, 6, 6, 6, 6, 8, 8, ~
## $ trans        <chr> "auto(l5)", "manual(m5)", "manual(m6)", "auto(av)", "auto~
## $ drv          <chr> "f", "f", "f", "f", "f", "f", "f", "4", "4", "4", "4", "4~
## $ cty          <int> 18, 21, 20, 21, 16, 18, 18, 18, 16, 20, 19, 15, 17, 17, 1~
## $ hwy          <int> 29, 29, 31, 30, 26, 26, 27, 26, 25, 28, 27, 25, 25, 25, 2~
## $ fl           <chr> "p", "p", "p", "p", "p", "p", "p", "p", "p", "p", "p", "p~
## $ class        <chr> "compact", "compact", "compact", "compact", "compact", "c~
```

```
dados <- mutate(.data = dados,
                across(where(is.character),
                as.factor))
#View(df)
glimpse(dados)
```

```
## Rows: 234
## Columns: 11
## $ manufacturer <fct> audi, audi, audi, audi, audi, audi, audi, audi, audi, aud~
## $ model        <fct> a4, a4, a4, a4, a4, a4, a4, a4 quattro, a4 quattro, a4 qu~
## $ displ        <dbl> 1.8, 1.8, 2.0, 2.0, 2.8, 2.8, 3.1, 1.8, 1.8, 2.0, 2.0, 2.~
## $ year         <int> 1999, 1999, 2008, 2008, 1999, 1999, 2008, 1999, 1999, 200~
## $ cyl          <int> 4, 4, 4, 4, 6, 6, 6, 4, 4, 4, 4, 6, 6, 6, 6, 6, 6, 8, 8, ~
## $ trans        <fct> auto(l5), manual(m5), manual(m6), auto(av), auto(l5), man~
## $ drv          <fct> f, f, f, f, f, f, f, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, r, ~
## $ cty          <int> 18, 21, 20, 21, 16, 18, 18, 18, 16, 20, 19, 15, 17, 17, 1~
## $ hwy          <int> 29, 29, 31, 30, 26, 26, 27, 26, 25, 28, 27, 25, 25, 25, 2~
## $ fl           <fct> p, p, p, p, p, p, p, p, p, p, p, p, p, p, p, p, p, p, r, ~
## $ class        <fct> compact, compact, compact, compact, compact, compact, com~
```

1.4 Propaganda 1 (Gustavo Jun Yakushiji)

Assistir os 3 vídeos do **Gustavo Jun Yakushiji** e **Cristian Villegas** no *youtube*

Dia 1 Introdução à Ciência de Dados (Introdução ao R e importação de dados)

Dia 2 Introdução à Ciência de Dados (Introdução ao ggplot2)

Dia 3 Introdução à Ciência de Dados (Introdução ao Tidyverse)

1.5 Propaganda 2 (Gustavo Jun Yakushiji)

Introdução à Ciência de Dados em R, Gustavo Jun Yakushiji; Cristian Villegas

1.6 Propaganda 3

Como pegar o código fonte do meu github? Link do meu github

Como pegar o arquivo PDF da apresentação? link: Um breve tutorial sobre dplyr com ggplot2 ou Arquivo PDF

Capítulo 2

dplyr (60 minutos)

2.1 Lista de funções do pacote dplyr

```
ls("package:dplyr")
```

```
##   [1] "%>%"                  "across"                "add_count"
##   [4] "add_count_"           "add_row"               "add_rownames"
##   [7] "add_tally"            "add_tally_"            "all_equal"
##  [10] "all_of"               "all_vars"              "anti_join"
##  [13] "any_of"               "any_vars"              "arrange"
##  [16] "arrange_"             "arrange_all"           "arrange_at"
##  [19] "arrange_if"           "as.tbl"                "as_data_frame"
##  [22] "as_label"             "as_tibble"             "auto_copy"
##  [25] "band_instruments"     "band_instruments2"     "band_members"
##  [28] "bench_tbls"           "between"               "bind_cols"
##  [31] "bind_rows"            "c_across"              "case_when"
##  [34] "changes"              "check_dbplyr"          "coalesce"
##  [37] "collapse"             "collect"               "combine"
##  [40] "common_by"            "compare_tbls"          "compare_tbls2"
##  [43] "compute"              "contains"              "copy_to"
##  [46] "count"                "count_"                "cumall"
##  [49] "cumany"               "cume_dist"             "cummean"
##  [52] "cur_column"           "cur_data"              "cur_data_all"
##  [55] "cur_group"            "cur_group_id"          "cur_group_rows"
##  [58] "current_vars"         "data_frame"            "data_frame_"
##  [61] "db_analyze"           "db_begin"              "db_commit"
##  [64] "db_create_index"      "db_create_indexes"     "db_create_table"
##  [67] "db_data_type"         "db_desc"               "db_drop_table"
```

```
##  [70] "db_explain"            "db_has_table"          "db_insert_into"
##  [73] "db_list_tables"        "db_query_fields"       "db_query_rows"
##  [76] "db_rollback"           "db_save_query"         "db_write_table"
##  [79] "dense_rank"            "desc"                  "dim_desc"
##  [82] "distinct"              "distinct_"             "distinct_all"
##  [85] "distinct_at"           "distinct_if"           "distinct_prepare"
##  [88] "do"                    "do_"                   "dplyr_col_modify"
##  [91] "dplyr_reconstruct"     "dplyr_row_slice"       "ends_with"
##  [94] "enexpr"                "enexprs"               "enquo"
##  [97] "enquos"                "ensym"                 "ensyms"
## [100] "eval_tbls"             "eval_tbls2"            "everything"
## [103] "explain"               "expr"                  "failwith"
## [106] "filter"                "filter_"               "filter_all"
## [109] "filter_at"             "filter_if"             "first"
## [112] "frame_data"            "full_join"             "funs"
## [115] "funs_"                 "glimpse"               "group_by"
## [118] "group_by_"             "group_by_all"          "group_by_at"
## [121] "group_by_drop_default" "group_by_if"           "group_by_prepare"
## [124] "group_cols"            "group_data"            "group_indices"
## [127] "group_indices_"        "group_keys"            "group_map"
## [130] "group_modify"          "group_nest"            "group_rows"
## [133] "group_size"            "group_split"           "group_trim"
## [136] "group_vars"            "group_walk"            "grouped_df"
## [139] "groups"                "id"                    "ident"
## [142] "if_all"                "if_any"                "if_else"
## [145] "inner_join"            "intersect"             "is.grouped_df"
## [148] "is.src"                "is.tbl"                "is_grouped_df"
## [151] "lag"                   "last"                  "last_col"
## [154] "lead"                  "left_join"             "location"
## [157] "lst"                   "lst_"                  "make_tbl"
## [160] "matches"               "min_rank"              "mutate"
## [163] "mutate_"               "mutate_all"            "mutate_at"
## [166] "mutate_each"           "mutate_each_"          "mutate_if"
## [169] "n"                     "n_distinct"            "n_groups"
## [172] "na_if"                 "near"                  "nest_by"
## [175] "nest_join"             "new_grouped_df"        "new_rowwise_df"
## [178] "nth"                   "ntile"                 "num_range"
## [181] "one_of"                "order_by"              "percent_rank"
## [184] "progress_estimated"    "pull"                  "quo"
## [187] "quo_name"              "quos"                  "recode"
## [190] "recode_factor"         "relocate"              "rename"
## [193] "rename_"               "rename_all"            "rename_at"
## [196] "rename_if"             "rename_vars"           "rename_vars_"
## [199] "rename_with"           "right_join"            "row_number"
## [202] "rows_append"           "rows_delete"           "rows_insert"
## [205] "rows_patch"            "rows_update"           "rows_upsert"
```

```
## [208] "rowwise"              "same_src"              "sample_frac"
## [211] "sample_n"             "select"                "select_"
## [214] "select_all"           "select_at"             "select_if"
## [217] "select_var"           "select_vars"           "select_vars_"
## [220] "semi_join"            "setdiff"               "setequal"
## [223] "show_query"           "slice"                 "slice_"
## [226] "slice_head"           "slice_max"             "slice_min"
## [229] "slice_sample"         "slice_tail"            "sql"
## [232] "sql_escape_ident"     "sql_escape_string"     "sql_join"
## [235] "sql_select"           "sql_semi_join"         "sql_set_op"
## [238] "sql_subquery"         "sql_translate_env"     "src"
## [241] "src_df"               "src_local"             "src_mysql"
## [244] "src_postgres"         "src_sqlite"            "src_tbls"
## [247] "starts_with"          "starwars"              "storms"
## [250] "summarise"            "summarise_"            "summarise_all"
## [253] "summarise_at"         "summarise_each"        "summarise_each_"
## [256] "summarise_if"         "summarize"             "summarize_"
## [259] "summarize_all"        "summarize_at"          "summarize_each"
## [262] "summarize_each_"      "summarize_if"          "sym"
## [265] "syms"                 "tally"                 "tally_"
## [268] "tbl"                  "tbl_df"                "tbl_nongroup_vars"
## [271] "tbl_ptype"            "tbl_sum"               "tbl_vars"
## [274] "tibble"               "top_frac"              "top_n"
## [277] "transmute"            "transmute_"            "transmute_all"
## [280] "transmute_at"         "transmute_if"          "tribble"
## [283] "type_sum"             "ungroup"               "union"
## [286] "union_all"            "validate_grouped_df"   "validate_rowwise_df"
## [289] "vars"                 "with_groups"           "with_order"
## [292] "wrap_dbplyr_obj"
```

2.2 Operador Pipe

```
sqrt(log(44))
```

```
## [1] 1.945299
```

```
44 %>% log %>% sqrt
```

```
## [1] 1.945299
```

2.3 select() para colunas

```
select(dados, manufacturer, model, year)
```

```
## # A tibble: 234 x 3
##    manufacturer model       year
##    <fct>        <fct>      <int>
##  1 audi         a4          1999
##  2 audi         a4          1999
##  3 audi         a4          2008
##  4 audi         a4          2008
##  5 audi         a4          1999
##  6 audi         a4          1999
##  7 audi         a4          2008
##  8 audi         a4 quattro  1999
##  9 audi         a4 quattro  1999
## 10 audi         a4 quattro  2008
## # ... with 224 more rows
```

```
select(dados, starts_with("m"))
```

```
## # A tibble: 234 x 2
##    manufacturer model
##    <fct>        <fct>
##  1 audi         a4
##  2 audi         a4
##  3 audi         a4
##  4 audi         a4
##  5 audi         a4
##  6 audi         a4
##  7 audi         a4
##  8 audi         a4 quattro
##  9 audi         a4 quattro
## 10 audi         a4 quattro
## # ... with 224 more rows
```

```
select(dados, contains("r"))
```

```
## # A tibble: 234 x 4
##    manufacturer  year trans      drv
##    <fct>        <int> <fct>      <fct>
##  1 audi          1999 auto(l5)   f
```

```
##  2 audi          1999 manual(m5) f
##  3 audi          2008 manual(m6) f
##  4 audi          2008 auto(av)   f
##  5 audi          1999 auto(l5)   f
##  6 audi          1999 manual(m5) f
##  7 audi          2008 auto(av)   f
##  8 audi          1999 manual(m5) 4
##  9 audi          1999 auto(l5)   4
## 10 audi          2008 manual(m6) 4
## # ... with 224 more rows
```

```
select(dados, ends_with("y"))
```

```
## # A tibble: 234 x 2
##      cty   hwy
##    <int> <int>
##  1    18    29
##  2    21    29
##  3    20    31
##  4    21    30
##  5    16    26
##  6    18    26
##  7    18    27
##  8    18    26
##  9    16    25
## 10    20    28
## # ... with 224 more rows
```

```
select(dados, matches("[abc]"))
```

```
## # A tibble: 234 x 6
##    manufacturer  year   cyl trans        cty class
##    <fct>        <int> <int> <fct>      <int> <fct>
##  1 audi          1999     4 auto(l5)      18 compact
##  2 audi          1999     4 manual(m5)    21 compact
##  3 audi          2008     4 manual(m6)    20 compact
##  4 audi          2008     4 auto(av)      21 compact
##  5 audi          1999     6 auto(l5)      16 compact
##  6 audi          1999     6 manual(m5)    18 compact
##  7 audi          2008     6 auto(av)      18 compact
##  8 audi          1999     4 manual(m5)    18 compact
##  9 audi          1999     4 auto(l5)      16 compact
## 10 audi          2008     4 manual(m6)    20 compact
## # ... with 224 more rows
```

```r
select(dados, starts_with("m"), starts_with("c"))
```

```
## # A tibble: 234 x 5
##    manufacturer model        cyl   cty class
##    <fct>        <fct>      <int> <int> <fct>
##  1 audi         a4             4    18 compact
##  2 audi         a4             4    21 compact
##  3 audi         a4             4    20 compact
##  4 audi         a4             4    21 compact
##  5 audi         a4             6    16 compact
##  6 audi         a4             6    18 compact
##  7 audi         a4             6    18 compact
##  8 audi         a4 quattro     4    18 compact
##  9 audi         a4 quattro     4    16 compact
## 10 audi         a4 quattro     4    20 compact
## # ... with 224 more rows
```

```r
select(dados, ends_with("l"), ends_with("s"))
```

```
## # A tibble: 234 x 6
##    model      displ   cyl fl    trans      class
##    <fct>      <dbl> <int> <fct> <fct>      <fct>
##  1 a4           1.8     4 p     auto(l5)   compact
##  2 a4           1.8     4 p     manual(m5) compact
##  3 a4           2       4 p     manual(m6) compact
##  4 a4           2       4 p     auto(av)   compact
##  5 a4           2.8     6 p     auto(l5)   compact
##  6 a4           2.8     6 p     manual(m5) compact
##  7 a4           3.1     6 p     auto(av)   compact
##  8 a4 quattro   1.8     4 p     manual(m5) compact
##  9 a4 quattro   1.8     4 p     auto(l5)   compact
## 10 a4 quattro   2       4 p     manual(m6) compact
## # ... with 224 more rows
```

```r
select(dados, 1:3)
```

```
## # A tibble: 234 x 3
##    manufacturer model      displ
##    <fct>        <fct>      <dbl>
##  1 audi         a4           1.8
##  2 audi         a4           1.8
##  3 audi         a4           2
##  4 audi         a4           2
```

```
##  5 audi         a4           2.8
##  6 audi         a4           2.8
##  7 audi         a4           3.1
##  8 audi         a4 quattro   1.8
##  9 audi         a4 quattro   1.8
## 10 audi         a4 quattro   2
## # ... with 224 more rows
```

```
select(dados, c(2,5,7))
```

```
## # A tibble: 234 x 3
##    model        cyl drv
##    <fct>      <int> <fct>
##  1 a4             4 f
##  2 a4             4 f
##  3 a4             4 f
##  4 a4             4 f
##  5 a4             6 f
##  6 a4             6 f
##  7 a4             6 f
##  8 a4 quattro     4 4
##  9 a4 quattro     4 4
## 10 a4 quattro     4 4
## # ... with 224 more rows
```

```
select(dados, manufacturer:cyl)
```

```
## # A tibble: 234 x 5
##    manufacturer model      displ  year   cyl
##    <fct>        <fct>      <dbl> <int> <int>
##  1 audi         a4           1.8  1999     4
##  2 audi         a4           1.8  1999     4
##  3 audi         a4           2    2008     4
##  4 audi         a4           2    2008     4
##  5 audi         a4           2.8  1999     6
##  6 audi         a4           2.8  1999     6
##  7 audi         a4           3.1  2008     6
##  8 audi         a4 quattro   1.8  1999     4
##  9 audi         a4 quattro   1.8  1999     4
## 10 audi         a4 quattro   2    2008     4
## # ... with 224 more rows
```

```
select(dados,-(manufacturer:cyl))
```

```
## # A tibble: 234 x 6
##    trans      drv     cty   hwy fl    class
##    <fct>      <fct> <int> <int> <fct> <fct>
##  1 auto(l5)   f        18    29 p     compact
##  2 manual(m5) f        21    29 p     compact
##  3 manual(m6) f        20    31 p     compact
##  4 auto(av)   f        21    30 p     compact
##  5 auto(l5)   f        16    26 p     compact
##  6 manual(m5) f        18    26 p     compact
##  7 auto(av)   f        18    27 p     compact
##  8 manual(m5) 4        18    26 p     compact
##  9 auto(l5)   4        16    25 p     compact
## 10 manual(m6) 4        20    28 p     compact
## # ... with 224 more rows
```

2.4 rename()

```
dados1 <- rename(dados,
              mnfc = manufacturer,
              mod = model)
dados1
```

```
## # A tibble: 234 x 11
##    mnfc  mod        displ  year   cyl trans      drv     cty   hwy fl    class
##    <fct> <fct>      <dbl> <int> <int> <fct>      <fct> <int> <int> <fct> <fct>
##  1 audi  a4           1.8  1999     4 auto(l5)   f        18    29 p     compact
##  2 audi  a4           1.8  1999     4 manual(m5) f        21    29 p     compact
##  3 audi  a4           2    2008     4 manual(m6) f        20    31 p     compact
##  4 audi  a4           2    2008     4 auto(av)   f        21    30 p     compact
##  5 audi  a4           2.8  1999     6 auto(l5)   f        16    26 p     compact
##  6 audi  a4           2.8  1999     6 manual(m5) f        18    26 p     compact
##  7 audi  a4           3.1  2008     6 auto(av)   f        18    27 p     compact
##  8 audi  a4 quattro   1.8  1999     4 manual(m5) 4        18    26 p     compact
##  9 audi  a4 quattro   1.8  1999     4 auto(l5)   4        16    25 p     compact
## 10 audi  a4 quattro   2    2008     4 manual(m6) 4        20    28 p     compact
## # ... with 224 more rows
```

```
select(dados,
       mnfc = manufacturer,
       mod = model)
```

```
## # A tibble: 234 x 2
##    mnfc  mod
##    <fct> <fct>
##  1 audi  a4
##  2 audi  a4
##  3 audi  a4
##  4 audi  a4
##  5 audi  a4
##  6 audi  a4
##  7 audi  a4
##  8 audi  a4 quattro
##  9 audi  a4 quattro
## 10 audi  a4 quattro
## # ... with 224 more rows
```

```
select(dados,
       mnfc = manufacturer,
       mod = model,
       everything())
```

```
## # A tibble: 234 x 11
##    mnfc  mod        displ  year   cyl trans      drv     cty   hwy fl    class
##    <fct> <fct>      <dbl> <int> <int> <fct>      <fct> <int> <int> <fct> <fct>
##  1 audi  a4           1.8  1999     4 auto(l5)   f        18    29 p     compact
##  2 audi  a4           1.8  1999     4 manual(m5) f        21    29 p     compact
##  3 audi  a4           2    2008     4 manual(m6) f        20    31 p     compact
##  4 audi  a4           2    2008     4 auto(av)   f        21    30 p     compact
##  5 audi  a4           2.8  1999     6 auto(l5)   f        16    26 p     compact
##  6 audi  a4           2.8  1999     6 manual(m5) f        18    26 p     compact
##  7 audi  a4           3.1  2008     6 auto(av)   f        18    27 p     compact
##  8 audi  a4 quattro   1.8  1999     4 manual(m5) 4        18    26 p     compact
##  9 audi  a4 quattro   1.8  1999     4 auto(l5)   4        16    25 p     compact
## 10 audi  a4 quattro   2    2008     4 manual(m6) 4        20    28 p     compact
## # ... with 224 more rows
```

2.5 mutate() para colunas

```
dados <- mutate(dados,
             `avg miles per gallon` = (cty + hwy) / 2)
dados
```

```
## # A tibble: 234 x 12
```

```
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  2 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
##  3 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  4 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  5 audi      a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
##  6 audi      a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
##  7 audi      a4      3.1  2008     6 auto~ f        18    27 p     comp~    22.5
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 audi      a4 q~   2    2008     4 manu~ 4        20    28 p     comp~    24
## # ... with 224 more rows, and abbreviated variable names 1: manufacturer,
## #   2: `avg miles per gallon`
```

```
dados <- mutate(dados,
             car = paste(manufacturer, model, sep = " "),
             `cyl / trans` = paste(cyl, " cylinders", " / ", trans, " transmission", se
dados
```

```
## # A tibble: 234 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  2 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
##  3 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  4 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  5 audi      a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
##  6 audi      a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
##  7 audi      a4      3.1  2008     6 auto~ f        18    27 p     comp~    22.5
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 audi      a4 q~   2    2008     4 manu~ 4        20    28 p     comp~    24
## # ... with 224 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

2.6 transmute()

```
transmute(dados,
          `avg miles per gallon` = (cty + hwy) / 2)
```

```
## # A tibble: 234 x 1
```

```
##    `avg miles per gallon`
##                     <dbl>
##  1                   23.5
##  2                   25
##  3                   25.5
##  4                   25.5
##  5                   21
##  6                   22
##  7                   22.5
##  8                   22
##  9                   20.5
## 10                   24
## # ... with 224 more rows
```

```
transmute(dados,
          car = paste(manufacturer, model, sep = " "),
          `cyl / trans` = paste(cyl, " cylinders", " / ", trans, " transmission", sep = ""))
```

```
## # A tibble: 234 x 2
##    car             `cyl / trans`
##    <chr>           <chr>
##  1 audi a4         4 cylinders / auto(l5) transmission
##  2 audi a4         4 cylinders / manual(m5) transmission
##  3 audi a4         4 cylinders / manual(m6) transmission
##  4 audi a4         4 cylinders / auto(av) transmission
##  5 audi a4         6 cylinders / auto(l5) transmission
##  6 audi a4         6 cylinders / manual(m5) transmission
##  7 audi a4         6 cylinders / auto(av) transmission
##  8 audi a4 quattro 4 cylinders / manual(m5) transmission
##  9 audi a4 quattro 4 cylinders / auto(l5) transmission
## 10 audi a4 quattro 4 cylinders / manual(m6) transmission
## # ... with 224 more rows
```

2.7 filter() para linhas

```
filter(dados, manufacturer == "audi")
```

```
## # A tibble: 18 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  2 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
```

```
##  3 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  4 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  5 audi      a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
##  6 audi      a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
##  7 audi      a4      3.1  2008     6 auto~ f        18    27 p     comp~    22.5
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 audi      a4 q~   2    2008     4 manu~ 4        20    28 p     comp~    24
## 11 audi      a4 q~   2    2008     4 auto~ 4        19    27 p     comp~    23
## 12 audi      a4 q~   2.8  1999     6 auto~ 4        15    25 p     comp~    20
## 13 audi      a4 q~   2.8  1999     6 manu~ 4        17    25 p     comp~    21
## 14 audi      a4 q~   3.1  2008     6 auto~ 4        17    25 p     comp~    21
## 15 audi      a4 q~   3.1  2008     6 manu~ 4        15    25 p     comp~    20
## 16 audi      a6 q~   2.8  1999     6 auto~ 4        15    24 p     mids~    19.5
## 17 audi      a6 q~   3.1  2008     6 auto~ 4        17    25 p     mids~    21
## 18 audi      a6 q~   4.2  2008     8 auto~ 4        16    23 p     mids~    19.5
## # ... with 2 more variables: car <chr>, `cyl / trans` <chr>, and abbreviated
## #   variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
filter(dados, manufacturer == "audi" & year == 1999)
```

```
## # A tibble: 9 x 14
##   manufact~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##   <fct>      <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
## 1 audi       a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
## 2 audi       a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
## 3 audi       a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
## 4 audi       a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
## 5 audi       a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
## 6 audi       a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 7 audi       a4 q~   2.8  1999     6 auto~ 4        15    25 p     comp~    20
## 8 audi       a4 q~   2.8  1999     6 manu~ 4        17    25 p     comp~    21
## 9 audi       a6 q~   2.8  1999     6 auto~ 4        15    24 p     mids~    19.5
## # ... with 2 more variables: car <chr>, `cyl / trans` <chr>, and abbreviated
## #   variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
filter(dados, manufacturer == "audi", year == 1999)
```

```
## # A tibble: 9 x 14
##   manufact~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##   <fct>      <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
## 1 audi       a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
## 2 audi       a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
## 3 audi       a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
```

```
## 4 audi       a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
## 5 audi       a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
## 6 audi       a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 7 audi       a4 q~   2.8  1999     6 auto~ 4        15    25 p     comp~    20
## 8 audi       a4 q~   2.8  1999     6 manu~ 4        17    25 p     comp~    21
## 9 audi       a6 q~   2.8  1999     6 auto~ 4        15    24 p     mids~    19.5
## # ... with 2 more variables: car <chr>, `cyl / trans` <chr>, and abbreviated
## #   variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
filter(dados, manufacturer == "audi" | manufacturer == "dodge")
```

```
## # A tibble: 55 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  2 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
##  3 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  4 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  5 audi      a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
##  6 audi      a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
##  7 audi      a4      3.1  2008     6 auto~ f        18    27 p     comp~    22.5
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 audi      a4 q~   2    2008     4 manu~ 4        20    28 p     comp~    24
## # ... with 45 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
filter(dados, manufacturer %in% c("audi", "dodge"))
```

```
## # A tibble: 55 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  2 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
##  3 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  4 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  5 audi      a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
##  6 audi      a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
##  7 audi      a4      3.1  2008     6 auto~ f        18    27 p     comp~    22.5
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 audi      a4 q~   2    2008     4 manu~ 4        20    28 p     comp~    24
## # ... with 45 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
filter(dados, hwy >= 30)
```

```
## # A tibble: 26 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  2 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  3 chevrolet mali~   2.4  2008     4 auto~ f        22    30 r     mids~    26
##  4 honda     civic   1.6  1999     4 manu~ f        28    33 r     subc~    30.5
##  5 honda     civic   1.6  1999     4 auto~ f        24    32 r     subc~    28
##  6 honda     civic   1.6  1999     4 manu~ f        25    32 r     subc~    28.5
##  7 honda     civic   1.6  1999     4 auto~ f        24    32 r     subc~    28
##  8 honda     civic   1.8  2008     4 manu~ f        26    34 r     subc~    30
##  9 honda     civic   1.8  2008     4 auto~ f        25    36 r     subc~    30.5
## 10 honda     civic   1.8  2008     4 auto~ f        24    36 c     subc~    30
## # ... with 16 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
filter(dados, year != 1999)
```

```
## # A tibble: 117 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  2 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  3 audi      a4      3.1  2008     6 auto~ f        18    27 p     comp~    22.5
##  4 audi      a4 q~   2    2008     4 manu~ 4        20    28 p     comp~    24
##  5 audi      a4 q~   2    2008     4 auto~ 4        19    27 p     comp~    23
##  6 audi      a4 q~   3.1  2008     6 auto~ 4        17    25 p     comp~    21
##  7 audi      a4 q~   3.1  2008     6 manu~ 4        15    25 p     comp~    20
##  8 audi      a6 q~   3.1  2008     6 auto~ 4        17    25 p     mids~    21
##  9 audi      a6 q~   4.2  2008     8 auto~ 4        16    23 p     mids~    19.5
## 10 chevrolet c150~   5.3  2008     8 auto~ r        14    20 r     suv      17
## # ... with 107 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
filter(dados, between(cty,15, 22))
```

```
## # A tibble: 143 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  2 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
```

```
##  3 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  4 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  5 audi      a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
##  6 audi      a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
##  7 audi      a4      3.1  2008     6 auto~ f        18    27 p     comp~    22.5
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 audi      a4 q~   2    2008     4 manu~ 4        20    28 p     comp~    24
## # ... with 133 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

2.8 slice() para linhas

```
slice(dados, 1:5)
```

```
## # A tibble: 5 x 14
##   manufact~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##   <fct>      <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
## 1 audi       a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
## 2 audi       a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
## 3 audi       a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
## 4 audi       a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
## 5 audi       a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
## # ... with 2 more variables: car <chr>, `cyl / trans` <chr>, and abbreviated
## #   variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
# dados[1:5,]

slice(dados, 20:30)
```

```
## # A tibble: 11 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 chevrolet c150~   5.3  2008     8 auto~ r        11    15 e     suv      13
##  2 chevrolet c150~   5.3  2008     8 auto~ r        14    20 r     suv      17
##  3 chevrolet c150~   5.7  1999     8 auto~ r        13    17 r     suv      15
##  4 chevrolet c150~   6    2008     8 auto~ r        12    17 r     suv      14.5
##  5 chevrolet corv~   5.7  1999     8 manu~ r        16    26 p     2sea~    21
##  6 chevrolet corv~   5.7  1999     8 auto~ r        15    23 p     2sea~    19
##  7 chevrolet corv~   6.2  2008     8 manu~ r        16    26 p     2sea~    21
##  8 chevrolet corv~   6.2  2008     8 auto~ r        15    25 p     2sea~    20
##  9 chevrolet corv~   7    2008     8 manu~ r        15    24 p     2sea~    19.5
```

```
## 10 chevrolet k150~   5.3  2008     8 auto~ 4        14    19 r     suv      16.5
## 11 chevrolet k150~   5.3  2008     8 auto~ 4        11    14 e     suv      12.5
## # ... with 2 more variables: car <chr>, `cyl / trans` <chr>, and abbreviated
## #   variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
# dados[20:30,]
```

2.9 arrange() para linhas

```
# ordenar "displ" de menor a maior
arrange(dados, displ)
```

```
## # A tibble: 234 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 honda     civic   1.6  1999     4 manu~ f        28    33 r     subc~    30.5
##  2 honda     civic   1.6  1999     4 auto~ f        24    32 r     subc~    28
##  3 honda     civic   1.6  1999     4 manu~ f        25    32 r     subc~    28.5
##  4 honda     civic   1.6  1999     4 manu~ f        23    29 p     subc~    26
##  5 honda     civic   1.6  1999     4 auto~ f        24    32 r     subc~    28
##  6 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  7 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 honda     civic   1.8  2008     4 manu~ f        26    34 r     subc~    30
## # ... with 224 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
arrange(dados, displ) %>%
  print(n=20)
```

```
## # A tibble: 234 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 honda     civic   1.6  1999     4 manu~ f        28    33 r     subc~    30.5
##  2 honda     civic   1.6  1999     4 auto~ f        24    32 r     subc~    28
##  3 honda     civic   1.6  1999     4 manu~ f        25    32 r     subc~    28.5
##  4 honda     civic   1.6  1999     4 manu~ f        23    29 p     subc~    26
##  5 honda     civic   1.6  1999     4 auto~ f        24    32 r     subc~    28
##  6 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  7 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
```

```
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 honda     civic   1.8  2008     4 manu~ f        26    34 r     subc~    30
## 11 honda     civic   1.8  2008     4 auto~ f        25    36 r     subc~    30.5
## 12 honda     civic   1.8  2008     4 auto~ f        24    36 c     subc~    30
## 13 toyota    coro~   1.8  1999     4 auto~ f        24    30 r     comp~    27
## 14 toyota    coro~   1.8  1999     4 auto~ f        24    33 r     comp~    28.5
## 15 toyota    coro~   1.8  1999     4 manu~ f        26    35 r     comp~    30.5
## 16 toyota    coro~   1.8  2008     4 manu~ f        28    37 r     comp~    32.5
## 17 toyota    coro~   1.8  2008     4 auto~ f        26    35 r     comp~    30.5
## 18 volkswag~ pass~   1.8  1999     4 manu~ f        21    29 p     mids~    25
## 19 volkswag~ pass~   1.8  1999     4 auto~ f        18    29 p     mids~    23.5
## 20 volkswag~ jetta   1.9  1999     4 manu~ f        33    44 d     comp~    38.5
## # ... with 214 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```r
# ordenar "displ" de maior a  menor
arrange(dados, desc(displ))
```

```
## # A tibble: 234 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 chevrolet corv~   7    2008     8 manu~ r        15    24 p     2sea~    19.5
##  2 chevrolet k150~   6.5  1999     8 auto~ 4        14    17 d     suv      15.5
##  3 chevrolet corv~   6.2  2008     8 manu~ r        16    26 p     2sea~    21
##  4 chevrolet corv~   6.2  2008     8 auto~ r        15    25 p     2sea~    20
##  5 jeep      gran~   6.1  2008     8 auto~ 4        11    14 p     suv      12.5
##  6 chevrolet c150~   6    2008     8 auto~ r        12    17 r     suv      14.5
##  7 dodge     dura~   5.9  1999     8 auto~ 4        11    15 r     suv      13
##  8 dodge     ram ~   5.9  1999     8 auto~ 4        11    15 r     pick~    13
##  9 chevrolet c150~   5.7  1999     8 auto~ r        13    17 r     suv      15
## 10 chevrolet corv~   5.7  1999     8 manu~ r        16    26 p     2sea~    21
## # ... with 224 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```r
arrange(dados, desc(displ)) %>%
  print(n=20)
```

```
## # A tibble: 234 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 chevrolet corv~   7    2008     8 manu~ r        15    24 p     2sea~    19.5
##  2 chevrolet k150~   6.5  1999     8 auto~ 4        14    17 d     suv      15.5
##  3 chevrolet corv~   6.2  2008     8 manu~ r        16    26 p     2sea~    21
```

```
##  4 chevrolet corv~   6.2  2008     8 auto~ r        15    25 p     2sea~    20
##  5 jeep      gran~   6.1  2008     8 auto~ 4        11    14 p     suv      12.5
##  6 chevrolet c150~   6    2008     8 auto~ r        12    17 r     suv      14.5
##  7 dodge     dura~   5.9  1999     8 auto~ 4        11    15 r     suv      13
##  8 dodge     ram ~   5.9  1999     8 auto~ 4        11    15 r     pick~    13
##  9 chevrolet c150~   5.7  1999     8 auto~ r        13    17 r     suv      15
## 10 chevrolet corv~   5.7  1999     8 manu~ r        16    26 p     2sea~    21
## 11 chevrolet corv~   5.7  1999     8 auto~ r        15    23 p     2sea~    19
## 12 chevrolet k150~   5.7  1999     8 auto~ 4        11    15 r     suv      13
## 13 dodge     dura~   5.7  2008     8 auto~ 4        13    18 r     suv      15.5
## 14 dodge     ram ~   5.7  2008     8 auto~ 4        13    17 r     pick~    15
## 15 jeep      gran~   5.7  2008     8 auto~ 4        13    18 r     suv      15.5
## 16 toyota    land~   5.7  2008     8 auto~ 4        13    18 r     suv      15.5
## 17 nissan    path~   5.6  2008     8 auto~ 4        12    18 p     suv      15
## 18 ford      expe~   5.4  1999     8 auto~ r        11    17 r     suv      14
## 19 ford      expe~   5.4  2008     8 auto~ r        12    18 r     suv      15
## 20 ford      f150~   5.4  1999     8 auto~ 4        11    15 r     pick~    13
## # ... with 214 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
select(dados, displ, cty) %>%
  arrange(displ, cty) %>%
  print(n = 20)
```

```
## # A tibble: 234 x 2
##    displ   cty
##    <dbl> <int>
##  1   1.6    23
##  2   1.6    24
##  3   1.6    24
##  4   1.6    25
##  5   1.6    28
##  6   1.8    16
##  7   1.8    18
##  8   1.8    18
##  9   1.8    18
## 10   1.8    21
## 11   1.8    21
## 12   1.8    24
## 13   1.8    24
## 14   1.8    24
## 15   1.8    25
## 16   1.8    26
## 17   1.8    26
## 18   1.8    26
```

```
## 19   1.8    28
## 20   1.9    29
## # ... with 214 more rows
```

2.10 distinct() para linhas

```r
dados_exemplo <- data.frame(id = 1:3,
                            name = c("John", "Max", "Julia"))
dados_exemplo
```

```
##   id  name
## 1  1  John
## 2  2   Max
## 3  3 Julia
```

```r
# bind_rows == rbind()
dados_exemplo<- bind_rows(dados_exemplo, slice(dados_exemplo, 2))
dados_exemplo
```

```
##   id  name
## 1  1  John
## 2  2   Max
## 3  3 Julia
## 4  2   Max
```

```r
distinct(dados_exemplo)
```

```
##   id  name
## 1  1  John
## 2  2   Max
## 3  3 Julia
```

```r
dados_exemplo2 <- data.frame(id = c(1,1,2),
                            name = c("John", "Max", "Julia"))
dados_exemplo2
```

```
##   id  name
## 1  1  John
## 2  1   Max
## 3  2 Julia
```

```
distinct(dados_exemplo2)
```

```
##   id  name
## 1  1  John
## 2  1   Max
## 3  2 Julia
```

```
dados_duplicados <- select(dados, manufacturer, model)
dados_duplicados
```

```
## # A tibble: 234 x 2
##    manufacturer model
##    <fct>        <fct>
##  1 audi         a4
##  2 audi         a4
##  3 audi         a4
##  4 audi         a4
##  5 audi         a4
##  6 audi         a4
##  7 audi         a4
##  8 audi         a4 quattro
##  9 audi         a4 quattro
## 10 audi         a4 quattro
## # ... with 224 more rows
```

```
dados_nao_duplicados <- distinct(dados_duplicados)
dados_nao_duplicados
```

```
## # A tibble: 38 x 2
##    manufacturer model
##    <fct>        <fct>
##  1 audi         a4
##  2 audi         a4 quattro
##  3 audi         a6 quattro
##  4 chevrolet    c1500 suburban 2wd
##  5 chevrolet    corvette
##  6 chevrolet    k1500 tahoe 4wd
##  7 chevrolet    malibu
##  8 dodge        caravan 2wd
##  9 dodge        dakota pickup 4wd
## 10 dodge        durango 4wd
## # ... with 28 more rows
```

2.11 summarise()

```
summarise(dados, `média hwy` = mean(hwy))
```

```
## # A tibble: 1 x 1
##   `média hwy`
##         <dbl>
## 1        23.4
```

```
summarise(dados,
          `num. de dados` = n(),
          `num. modelos` = n_distinct(model))
```

```
## # A tibble: 1 x 2
##   `num. de dados` `num. modelos`
##             <int>          <int>
## 1             234             38
```

```
summarise(dados,
          `mín. hwy` = min(hwy, na.rm = TRUE),
          `mín. cty` = min(cty, na.rm = TRUE),
          `máx. hwy` = max(hwy, na.rm = TRUE),
          `máx. cty` = max(cty, na.rm = TRUE))
```

```
## # A tibble: 1 x 4
##   `mín. hwy` `mín. cty` `máx. hwy` `máx. cty`
##        <int>      <int>      <int>      <int>
## 1         12          9         44         35
```

```
dados %>%
  summarise_at(c("hwy", "cty"), list(min, max), na.rm = TRUE)
```

```
## # A tibble: 1 x 4
##   hwy_fn1 cty_fn1 hwy_fn2 cty_fn2
##     <int>   <int>   <int>   <int>
## 1      12       9      44      35
```

```
dados %>%
  summarise_if(is.numeric, list(min, max), na.rm = TRUE)
```

```
## # A tibble: 1 x 12
##   displ_fn1 year_fn1 cyl_fn1 cty_fn1 hwy_fn1 avg miles~1 displ~2 year_~3 cyl_fn2
##       <dbl>    <int>   <int>   <int>   <int>       <dbl>   <dbl>   <int>   <int>
## 1       1.6     1999       4       9      12        10.5       7    2008       8
## # ... with 3 more variables: cty_fn2 <int>, hwy_fn2 <int>,
## #   `avg miles per gallon_fn2` <dbl>, and abbreviated variable names
## #   1: `avg miles per gallon_fn1`, 2: displ_fn2, 3: year_fn2
```

2.12 group_by()

```
group_by(dados, manufacturer)
```

```
## # A tibble: 234 x 14
## # Groups:   manufacturer [15]
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 audi      a4      1.8  1999     4 auto~ f        18    29 p     comp~    23.5
##  2 audi      a4      1.8  1999     4 manu~ f        21    29 p     comp~    25
##  3 audi      a4      2    2008     4 manu~ f        20    31 p     comp~    25.5
##  4 audi      a4      2    2008     4 auto~ f        21    30 p     comp~    25.5
##  5 audi      a4      2.8  1999     6 auto~ f        16    26 p     comp~    21
##  6 audi      a4      2.8  1999     6 manu~ f        18    26 p     comp~    22
##  7 audi      a4      3.1  2008     6 auto~ f        18    27 p     comp~    22.5
##  8 audi      a4 q~   1.8  1999     4 manu~ 4        18    26 p     comp~    22
##  9 audi      a4 q~   1.8  1999     4 auto~ 4        16    25 p     comp~    20.5
## 10 audi      a4 q~   2    2008     4 manu~ 4        20    28 p     comp~    24
## # ... with 224 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
dados %>%
  group_by(manufacturer) %>%
  summarise(`num. carros` = n())
```

```
## # A tibble: 15 x 2
##    manufacturer `num. carros`
##    <fct>                <int>
##  1 audi                    18
##  2 chevrolet               19
##  3 dodge                   37
##  4 ford                    25
##  5 honda                    9
##  6 hyundai                 14
```

```
##  7 jeep                    8
##  8 land rover              4
##  9 lincoln                 3
## 10 mercury                 4
## 11 nissan                 13
## 12 pontiac                 5
## 13 subaru                 14
## 14 toyota                 34
## 15 volkswagen             27
```

```
dados %>%
  group_by(model) %>%
  summarise(`média hwy` = mean(hwy),
            `min. hwy` = min(hwy),
            `max. hwy` = max(hwy))
```

```
## # A tibble: 38 x 4
##    model              `média hwy` `min. hwy` `max. hwy`
##    <fct>                    <dbl>      <int>      <int>
##  1 4runner 4wd               18.8         17         20
##  2 a4                        28.3         26         31
##  3 a4 quattro                25.8         25         28
##  4 a6 quattro                24           23         25
##  5 altima                    28.7         26         32
##  6 c1500 suburban 2wd        17.8         15         20
##  7 camry                     28.3         26         31
##  8 camry solara              28.1         26         31
##  9 caravan 2wd               22.4         17         24
## 10 civic                     32.6         29         36
## # ... with 28 more rows
```

2.13 count()

```
count(dados)
```

```
## # A tibble: 1 x 1
##       n
##   <int>
## 1   234
```

```r
dados %>%
  group_by(manufacturer) %>%
  count()
```

```
## # A tibble: 15 x 2
## # Groups:   manufacturer [15]
##    manufacturer     n
##    <fct>        <int>
##  1 audi            18
##  2 chevrolet       19
##  3 dodge           37
##  4 ford            25
##  5 honda            9
##  6 hyundai         14
##  7 jeep             8
##  8 land rover       4
##  9 lincoln          3
## 10 mercury          4
## 11 nissan          13
## 12 pontiac          5
## 13 subaru          14
## 14 toyota          34
## 15 volkswagen      27
```

```r
# Equivalente com o código anterior
dados %>%
  group_by(manufacturer) %>%
  summarise(cars = n())
```

```
## # A tibble: 15 x 2
##    manufacturer  cars
##    <fct>        <int>
##  1 audi            18
##  2 chevrolet       19
##  3 dodge           37
##  4 ford            25
##  5 honda            9
##  6 hyundai         14
##  7 jeep             8
##  8 land rover       4
##  9 lincoln          3
## 10 mercury          4
## 11 nissan          13
## 12 pontiac          5
```

```
## 13 subaru          14
## 14 toyota          34
## 15 volkswagen      27
```

2.14 sample_n()

```
set.seed(567)
sample_n(dados, size = 10, replace = F)
```

```
## # A tibble: 10 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 mercury   moun~   5    1999     8 auto~ 4        13    17 r     suv      15
##  2 chevrolet corv~   7    2008     8 manu~ r        15    24 p     2sea~    19.5
##  3 dodge     ram ~   4.7  2008     8 manu~ 4        12    16 r     pick~    14
##  4 toyota    land~   4.7  1999     8 auto~ 4        11    15 r     suv      13
##  5 volkswag~ jetta   2    1999     4 auto~ f        19    26 r     comp~    22.5
##  6 dodge     cara~   3.8  1999     6 auto~ f        15    21 r     mini~    18
##  7 honda     civic   1.8  2008     4 auto~ f        25    36 r     subc~    30.5
##  8 ford      must~   4.6  1999     8 auto~ r        15    21 r     subc~    18
##  9 chevrolet c150~   5.3  2008     8 auto~ r        14    20 r     suv      17
## 10 ford      expe~   5.4  1999     8 auto~ r        11    17 r     suv      14
## # ... with 2 more variables: car <chr>, `cyl / trans` <chr>, and abbreviated
## #   variable names 1: manufacturer, 2: `avg miles per gallon`
```

```
sample_n(dados, size = 10, replace = T)
```

```
## # A tibble: 10 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 chevrolet c150~   5.3  2008     8 auto~ r        11    15 e     suv      13
##  2 volkswag~ gti     2    2008     4 auto~ f        22    29 p     comp~    25.5
##  3 dodge     dako~   4.7  2008     8 auto~ 4        14    19 r     pick~    16.5
##  4 ford      expl~   4.6  2008     8 auto~ 4        13    19 r     suv      16
##  5 dodge     cara~   3.8  2008     6 auto~ f        16    23 r     mini~    19.5
##  6 chevrolet k150~   5.3  2008     8 auto~ 4        14    19 r     suv      16.5
##  7 dodge     dura~   5.2  1999     8 auto~ 4        11    16 r     suv      13.5
##  8 toyota    camry   2.4  2008     4 manu~ f        21    31 r     mids~    26
##  9 toyota    camry   3    1999     6 manu~ f        18    26 r     mids~    22
## 10 subaru    impr~   2.2  1999     4 auto~ 4        21    26 r     subc~    23.5
## # ... with 2 more variables: car <chr>, `cyl / trans` <chr>, and abbreviated
## #   variable names 1: manufacturer, 2: `avg miles per gallon`
```

2.15 sample_frac()

```
sample_frac(dados, size = 0.1, replace = F)
```

```
## # A tibble: 23 x 14
##    manufac~1 model displ  year   cyl trans drv     cty   hwy fl    class avg m~2
##    <fct>     <fct> <dbl> <int> <int> <fct> <fct> <int> <int> <fct> <fct>   <dbl>
##  1 toyota    coro~   1.8  2008     4 manu~ f        28    37 r     comp~    32.5
##  2 lincoln   navi~   5.4  1999     8 auto~ r        11    17 r     suv      14
##  3 honda     civic   1.6  1999     4 auto~ f        24    32 r     subc~    28
##  4 audi      a6 q~   2.8  1999     6 auto~ 4        15    24 p     mids~    19.5
##  5 nissan    path~   4    2008     6 auto~ 4        14    20 p     suv      17
##  6 toyota    camry   3.5  2008     6 auto~ f        19    28 r     mids~    23.5
##  7 subaru    impr~   2.5  2008     4 auto~ 4        20    25 p     comp~    22.5
##  8 toyota    toyo~   3.4  1999     6 auto~ 4        15    19 r     pick~    17
##  9 audi      a4 q~   3.1  2008     6 manu~ 4        15    25 p     comp~    20
## 10 toyota    coro~   1.8  1999     4 manu~ f        26    35 r     comp~    30.5
## # ... with 13 more rows, 2 more variables: car <chr>, `cyl / trans` <chr>, and
## #   abbreviated variable names 1: manufacturer, 2: `avg miles per gallon`
```

Capítulo 3

ggplot2 (60 minutos)

Alguns links

The R Graph Gallery

120 registered extensions available to explore

link 1: patchwork

link 2: patchwork

3.1 Lista de funções do pacote ggplot2

```
ls("package:ggplot2")
```

```
##   [1] "%+%"                          "%+replace%"
##   [3] "aes"                          "aes_"
##   [5] "aes_all"                      "aes_auto"
##   [7] "aes_q"                        "aes_string"
##   [9] "after_scale"                  "after_stat"
##  [11] "alpha"                        "annotate"
##  [13] "annotation_custom"            "annotation_logticks"
##  [15] "annotation_map"               "annotation_raster"
##  [17] "arrow"                        "as_label"
##  [19] "as_labeller"                  "autolayer"
##  [21] "autoplot"                     "AxisSecondary"
##  [23] "benchplot"                    "binned_scale"
##  [25] "borders"                      "calc_element"
##  [27] "combine_vars"                 "continuous_scale"
```

```
##  [29] "Coord"                      "coord_cartesian"
##  [31] "coord_equal"                "coord_fixed"
##  [33] "coord_flip"                 "coord_map"
##  [35] "coord_munch"                "coord_polar"
##  [37] "coord_quickmap"             "coord_sf"
##  [39] "coord_trans"                "CoordCartesian"
##  [41] "CoordFixed"                 "CoordFlip"
##  [43] "CoordMap"                   "CoordPolar"
##  [45] "CoordQuickmap"              "CoordSf"
##  [47] "CoordTrans"                 "cut_interval"
##  [49] "cut_number"                 "cut_width"
##  [51] "derive"                     "diamonds"
##  [53] "discrete_scale"             "draw_key_abline"
##  [55] "draw_key_blank"             "draw_key_boxplot"
##  [57] "draw_key_crossbar"          "draw_key_dotplot"
##  [59] "draw_key_label"             "draw_key_linerange"
##  [61] "draw_key_path"              "draw_key_point"
##  [63] "draw_key_pointrange"        "draw_key_polygon"
##  [65] "draw_key_rect"              "draw_key_smooth"
##  [67] "draw_key_text"              "draw_key_timeseries"
##  [69] "draw_key_vline"             "draw_key_vpath"
##  [71] "dup_axis"                   "economics"
##  [73] "economics_long"             "el_def"
##  [75] "element_blank"              "element_grob"
##  [77] "element_line"               "element_rect"
##  [79] "element_render"             "element_text"
##  [81] "enexpr"                     "enexprs"
##  [83] "enquo"                      "enquos"
##  [85] "ensym"                      "ensyms"
##  [87] "expand_limits"              "expand_scale"
##  [89] "expansion"                  "expr"
##  [91] "Facet"                      "facet_grid"
##  [93] "facet_null"                 "facet_wrap"
##  [95] "FacetGrid"                  "FacetNull"
##  [97] "FacetWrap"                  "faithfuld"
##  [99] "find_panel"                 "flip_data"
## [101] "flipped_names"              "fortify"
## [103] "Geom"                       "geom_abline"
## [105] "geom_area"                  "geom_bar"
## [107] "geom_bin_2d"                "geom_bin2d"
## [109] "geom_blank"                 "geom_boxplot"
## [111] "geom_col"                   "geom_contour"
## [113] "geom_contour_filled"        "geom_count"
## [115] "geom_crossbar"              "geom_curve"
## [117] "geom_density"               "geom_density_2d"
## [119] "geom_density_2d_filled"     "geom_density2d"
```

```
## [121] "geom_density2d_filled"      "geom_dotplot"
## [123] "geom_errorbar"              "geom_errorbarh"
## [125] "geom_freqpoly"              "geom_function"
## [127] "geom_hex"                   "geom_histogram"
## [129] "geom_hline"                 "geom_jitter"
## [131] "geom_label"                 "geom_line"
## [133] "geom_linerange"             "geom_map"
## [135] "geom_path"                  "geom_point"
## [137] "geom_pointrange"            "geom_polygon"
## [139] "geom_qq"                    "geom_qq_line"
## [141] "geom_quantile"              "geom_raster"
## [143] "geom_rect"                  "geom_ribbon"
## [145] "geom_rug"                   "geom_segment"
## [147] "geom_sf"                    "geom_sf_label"
## [149] "geom_sf_text"               "geom_smooth"
## [151] "geom_spoke"                 "geom_step"
## [153] "geom_text"                  "geom_tile"
## [155] "geom_violin"                "geom_vline"
## [157] "GeomAbline"                 "GeomAnnotationMap"
## [159] "GeomArea"                   "GeomBar"
## [161] "GeomBlank"                  "GeomBoxplot"
## [163] "GeomCol"                    "GeomContour"
## [165] "GeomContourFilled"          "GeomCrossbar"
## [167] "GeomCurve"                  "GeomCustomAnn"
## [169] "GeomDensity"                "GeomDensity2d"
## [171] "GeomDensity2dFilled"        "GeomDotplot"
## [173] "GeomErrorbar"               "GeomErrorbarh"
## [175] "GeomFunction"               "GeomHex"
## [177] "GeomHline"                  "GeomLabel"
## [179] "GeomLine"                   "GeomLinerange"
## [181] "GeomLogticks"               "GeomMap"
## [183] "GeomPath"                   "GeomPoint"
## [185] "GeomPointrange"             "GeomPolygon"
## [187] "GeomQuantile"               "GeomRaster"
## [189] "GeomRasterAnn"              "GeomRect"
## [191] "GeomRibbon"                 "GeomRug"
## [193] "GeomSegment"                "GeomSf"
## [195] "GeomSmooth"                 "GeomSpoke"
## [197] "GeomStep"                   "GeomText"
## [199] "GeomTile"                   "GeomViolin"
## [201] "GeomVline"                  "get_alt_text"
## [203] "get_element_tree"           "gg_dep"
## [205] "ggplot"                     "ggplot_add"
## [207] "ggplot_build"               "ggplot_gtable"
## [209] "ggplotGrob"                 "ggproto"
## [211] "ggproto_parent"             "ggsave"
```

```
## [213] "ggtitle"                    "guide_axis"
## [215] "guide_bins"                 "guide_colorbar"
## [217] "guide_colorsteps"           "guide_colourbar"
## [219] "guide_coloursteps"          "guide_gengrob"
## [221] "guide_geom"                 "guide_legend"
## [223] "guide_merge"                "guide_none"
## [225] "guide_train"                "guide_transform"
## [227] "guides"                     "has_flipped_aes"
## [229] "is.Coord"                   "is.facet"
## [231] "is.ggplot"                  "is.ggproto"
## [233] "is.theme"                   "label_both"
## [235] "label_bquote"               "label_context"
## [237] "label_parsed"               "label_value"
## [239] "label_wrap_gen"             "labeller"
## [241] "labs"                       "last_plot"
## [243] "layer"                      "layer_data"
## [245] "layer_grob"                 "layer_scales"
## [247] "layer_sf"                   "Layout"
## [249] "lims"                       "luv_colours"
## [251] "map_data"                   "margin"
## [253] "max_height"                 "max_width"
## [255] "mean_cl_boot"               "mean_cl_normal"
## [257] "mean_sdl"                   "mean_se"
## [259] "median_hilow"               "merge_element"
## [261] "midwest"                    "mpg"
## [263] "msleep"                     "panel_cols"
## [265] "panel_rows"                 "Position"
## [267] "position_dodge"             "position_dodge2"
## [269] "position_fill"              "position_identity"
## [271] "position_jitter"            "position_jitterdodge"
## [273] "position_nudge"             "position_stack"
## [275] "PositionDodge"              "PositionDodge2"
## [277] "PositionFill"               "PositionIdentity"
## [279] "PositionJitter"             "PositionJitterdodge"
## [281] "PositionNudge"              "PositionStack"
## [283] "presidential"               "qplot"
## [285] "quickplot"                  "quo"
## [287] "quo_name"                   "quos"
## [289] "register_theme_elements"    "rel"
## [291] "remove_missing"             "render_axes"
## [293] "render_strips"              "reset_theme_settings"
## [295] "resolution"                 "Scale"
## [297] "scale_alpha"                "scale_alpha_binned"
## [299] "scale_alpha_continuous"     "scale_alpha_date"
## [301] "scale_alpha_datetime"       "scale_alpha_discrete"
## [303] "scale_alpha_identity"       "scale_alpha_manual"
```

```
## [305] "scale_alpha_ordinal"        "scale_color_binned"
## [307] "scale_color_brewer"         "scale_color_continuous"
## [309] "scale_color_date"           "scale_color_datetime"
## [311] "scale_color_discrete"       "scale_color_distiller"
## [313] "scale_color_fermenter"      "scale_color_gradient"
## [315] "scale_color_gradient2"      "scale_color_gradientn"
## [317] "scale_color_grey"           "scale_color_hue"
## [319] "scale_color_identity"       "scale_color_manual"
## [321] "scale_color_ordinal"        "scale_color_steps"
## [323] "scale_color_steps2"         "scale_color_stepsn"
## [325] "scale_color_viridis_b"      "scale_color_viridis_c"
## [327] "scale_color_viridis_d"      "scale_colour_binned"
## [329] "scale_colour_brewer"        "scale_colour_continuous"
## [331] "scale_colour_date"          "scale_colour_datetime"
## [333] "scale_colour_discrete"      "scale_colour_distiller"
## [335] "scale_colour_fermenter"     "scale_colour_gradient"
## [337] "scale_colour_gradient2"     "scale_colour_gradientn"
## [339] "scale_colour_grey"          "scale_colour_hue"
## [341] "scale_colour_identity"      "scale_colour_manual"
## [343] "scale_colour_ordinal"       "scale_colour_steps"
## [345] "scale_colour_steps2"        "scale_colour_stepsn"
## [347] "scale_colour_viridis_b"     "scale_colour_viridis_c"
## [349] "scale_colour_viridis_d"     "scale_continuous_identity"
## [351] "scale_discrete_identity"    "scale_discrete_manual"
## [353] "scale_fill_binned"          "scale_fill_brewer"
## [355] "scale_fill_continuous"      "scale_fill_date"
## [357] "scale_fill_datetime"        "scale_fill_discrete"
## [359] "scale_fill_distiller"       "scale_fill_fermenter"
## [361] "scale_fill_gradient"        "scale_fill_gradient2"
## [363] "scale_fill_gradientn"       "scale_fill_grey"
## [365] "scale_fill_hue"             "scale_fill_identity"
## [367] "scale_fill_manual"          "scale_fill_ordinal"
## [369] "scale_fill_steps"           "scale_fill_steps2"
## [371] "scale_fill_stepsn"          "scale_fill_viridis_b"
## [373] "scale_fill_viridis_c"       "scale_fill_viridis_d"
## [375] "scale_linetype"             "scale_linetype_binned"
## [377] "scale_linetype_continuous"  "scale_linetype_discrete"
## [379] "scale_linetype_identity"    "scale_linetype_manual"
## [381] "scale_linewidth"            "scale_linewidth_binned"
## [383] "scale_linewidth_continuous" "scale_linewidth_date"
## [385] "scale_linewidth_datetime"   "scale_linewidth_discrete"
## [387] "scale_linewidth_ordinal"    "scale_radius"
## [389] "scale_shape"                "scale_shape_binned"
## [391] "scale_shape_continuous"     "scale_shape_discrete"
## [393] "scale_shape_identity"       "scale_shape_manual"
## [395] "scale_shape_ordinal"        "scale_size"
```

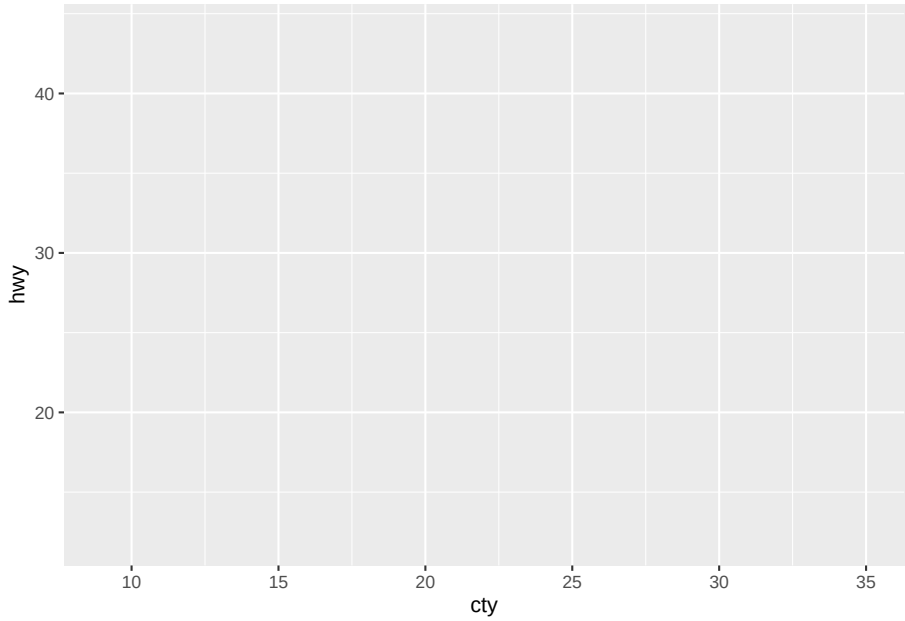
```
## [397] "scale_size_area"            "scale_size_binned"
## [399] "scale_size_binned_area"     "scale_size_continuous"
## [401] "scale_size_date"            "scale_size_datetime"
## [403] "scale_size_discrete"        "scale_size_identity"
## [405] "scale_size_manual"          "scale_size_ordinal"
## [407] "scale_type"                 "scale_x_binned"
## [409] "scale_x_continuous"         "scale_x_date"
## [411] "scale_x_datetime"           "scale_x_discrete"
## [413] "scale_x_log10"              "scale_x_reverse"
## [415] "scale_x_sqrt"               "scale_x_time"
## [417] "scale_y_binned"             "scale_y_continuous"
## [419] "scale_y_date"               "scale_y_datetime"
## [421] "scale_y_discrete"           "scale_y_log10"
## [423] "scale_y_reverse"            "scale_y_sqrt"
## [425] "scale_y_time"               "ScaleBinned"
## [427] "ScaleBinnedPosition"        "ScaleContinuous"
## [429] "ScaleContinuousDate"        "ScaleContinuousDatetime"
## [431] "ScaleContinuousIdentity"    "ScaleContinuousPosition"
## [433] "ScaleDiscrete"              "ScaleDiscreteIdentity"
## [435] "ScaleDiscretePosition"      "seals"
## [437] "sec_axis"                   "set_last_plot"
## [439] "sf_transform_xy"            "should_stop"
## [441] "stage"                      "standardise_aes_names"
## [443] "stat"                       "Stat"
## [445] "stat_align"                 "stat_bin"
## [447] "stat_bin_2d"                "stat_bin_hex"
## [449] "stat_bin2d"                 "stat_binhex"
## [451] "stat_boxplot"               "stat_contour"
## [453] "stat_contour_filled"        "stat_count"
## [455] "stat_density"               "stat_density_2d"
## [457] "stat_density_2d_filled"     "stat_density2d"
## [459] "stat_density2d_filled"      "stat_ecdf"
## [461] "stat_ellipse"               "stat_function"
## [463] "stat_identity"              "stat_qq"
## [465] "stat_qq_line"               "stat_quantile"
## [467] "stat_sf"                    "stat_sf_coordinates"
## [469] "stat_smooth"                "stat_spoke"
## [471] "stat_sum"                   "stat_summary"
## [473] "stat_summary_2d"            "stat_summary_bin"
## [475] "stat_summary_hex"           "stat_summary2d"
## [477] "stat_unique"                "stat_ydensity"
## [479] "StatAlign"                  "StatBin"
## [481] "StatBin2d"                  "StatBindot"
## [483] "StatBinhex"                 "StatBoxplot"
## [485] "StatContour"                "StatContourFilled"
## [487] "StatCount"                  "StatDensity"
```

```
## [489] "StatDensity2d"              "StatDensity2dFilled"
## [491] "StatEcdf"                   "StatEllipse"
## [493] "StatFunction"               "StatIdentity"
## [495] "StatQq"                     "StatQqLine"
## [497] "StatQuantile"               "StatSf"
## [499] "StatSfCoordinates"          "StatSmooth"
## [501] "StatSum"                    "StatSummary"
## [503] "StatSummary2d"              "StatSummaryBin"
## [505] "StatSummaryHex"             "StatUnique"
## [507] "StatYdensity"               "summarise_coord"
## [509] "summarise_layers"           "summarise_layout"
## [511] "sym"                        "syms"
## [513] "theme"                      "theme_bw"
## [515] "theme_classic"              "theme_dark"
## [517] "theme_get"                  "theme_gray"
## [519] "theme_grey"                 "theme_light"
## [521] "theme_linedraw"             "theme_minimal"
## [523] "theme_replace"              "theme_set"
## [525] "theme_test"                 "theme_update"
## [527] "theme_void"                 "transform_position"
## [529] "txhousing"                  "unit"
## [531] "update_geom_defaults"       "update_labels"
## [533] "update_stat_defaults"       "vars"
## [535] "waiver"                     "wrap_dims"
## [537] "xlab"                       "xlim"
## [539] "ylab"                       "ylim"
## [541] "zeroGrob"
```

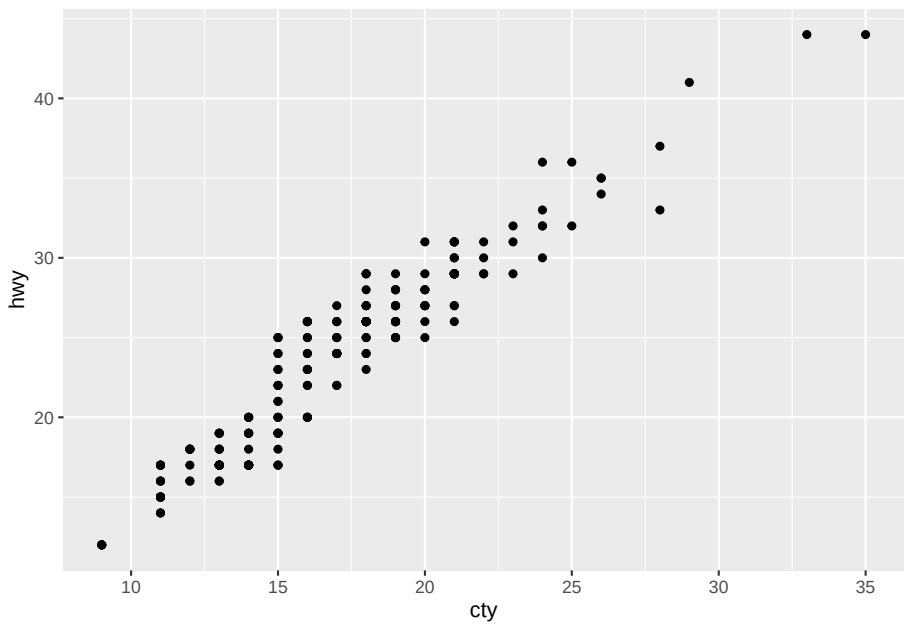
3.2 Primeiros passos usando geom_point

```
ggplot(dados)
```

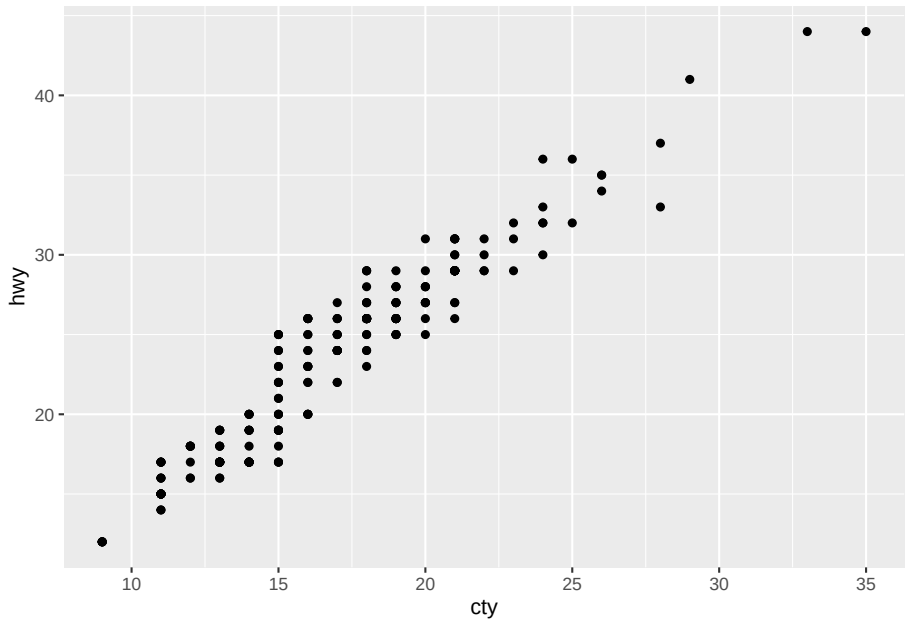
```
ggplot(dados, aes(x = cty, y = hwy))
```



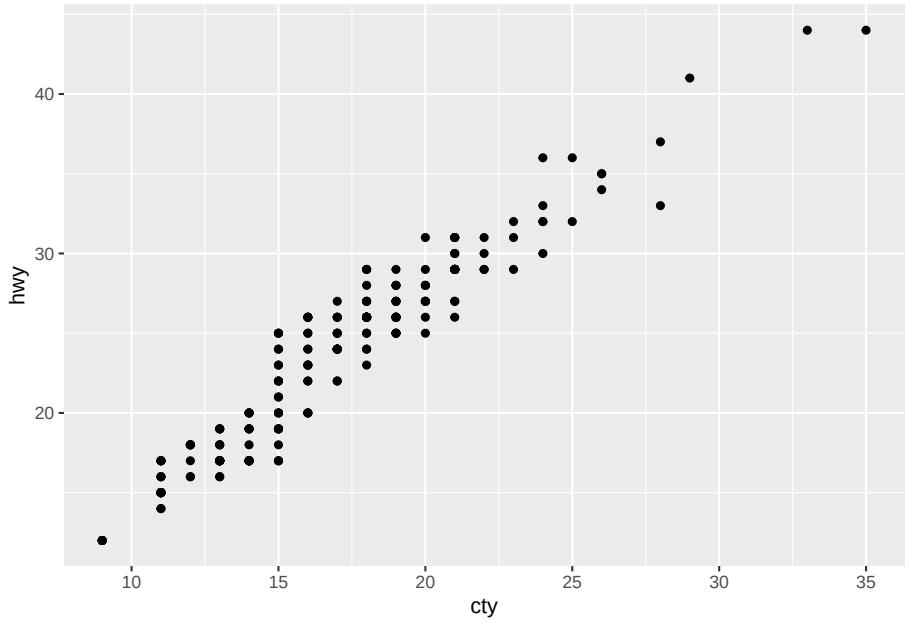
```
# Alternativas
ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point()
```



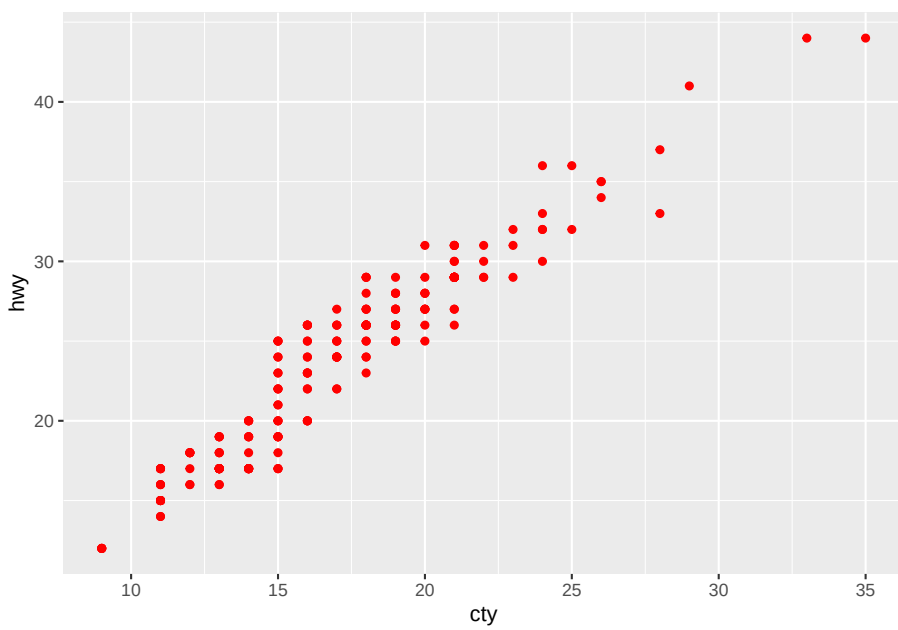

```
ggplot(dados) +
  geom_point(aes(x = cty, y = hwy))
```



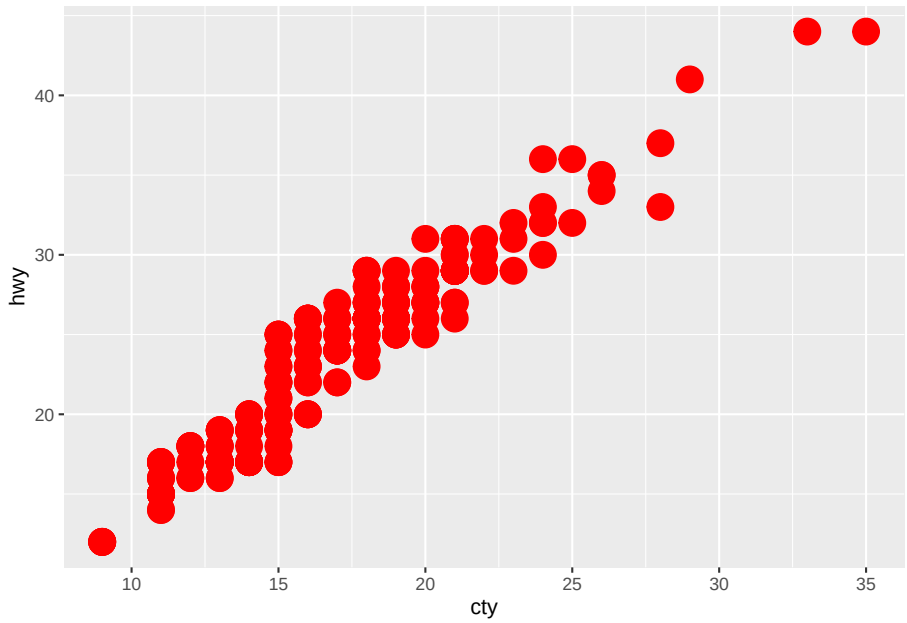

```
ggplot() +
  geom_point(data = dados, aes(x = cty, y = hwy))
```



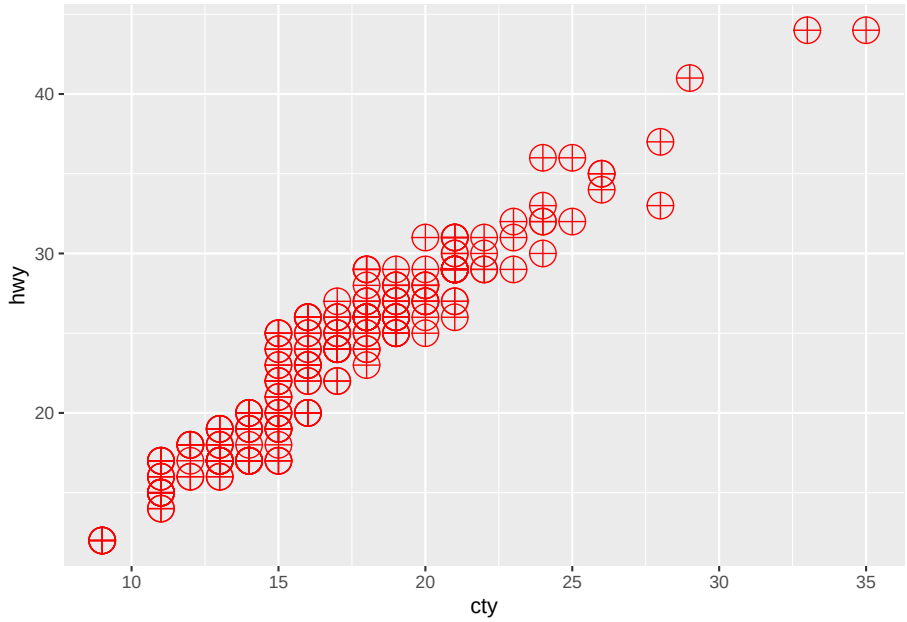
```
# Fim

ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point(colour = "red")
```



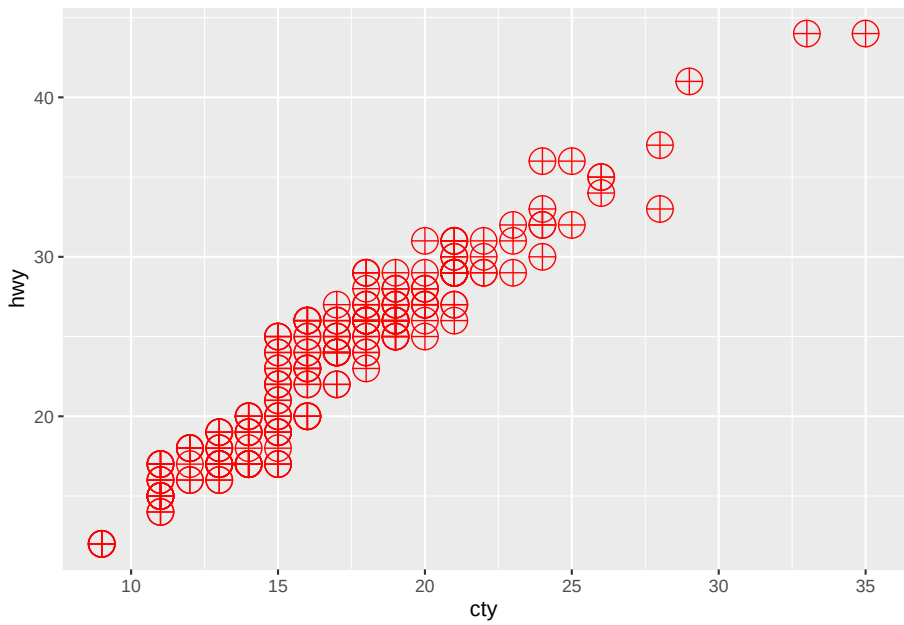

```
ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point(colour = "red", size = 6)
```



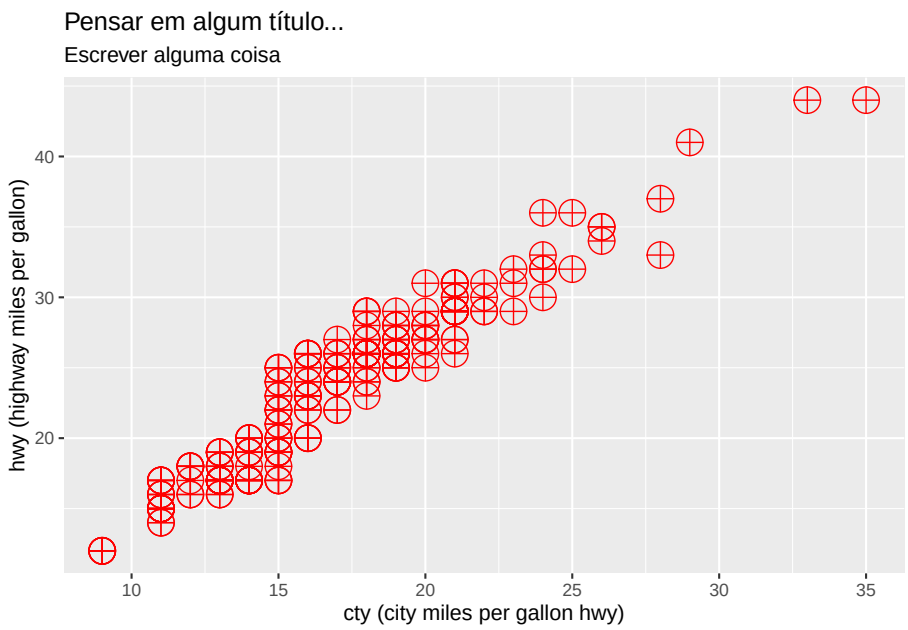

```
ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point(colour = "red", size = 6, shape = 10)
```



```
# Alternativa
ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point(colour = "red", size = 6, shape = "circle plus")
```




```
ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point(colour = "red", size = 6, shape = 10)+
  labs(x = "cty (city miles per gallon hwy)",
       y = "hwy (highway miles per gallon)",
       title = "Pensar em algum título...",
       subtitle = "Escrever alguma coisa")
```

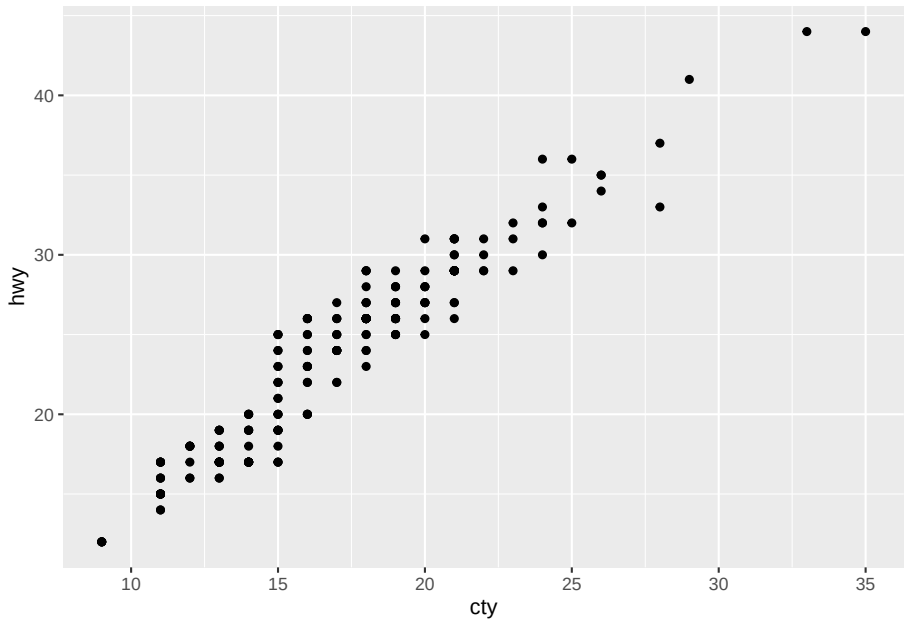


3.2.1 Mais detalhes sobre geom_point

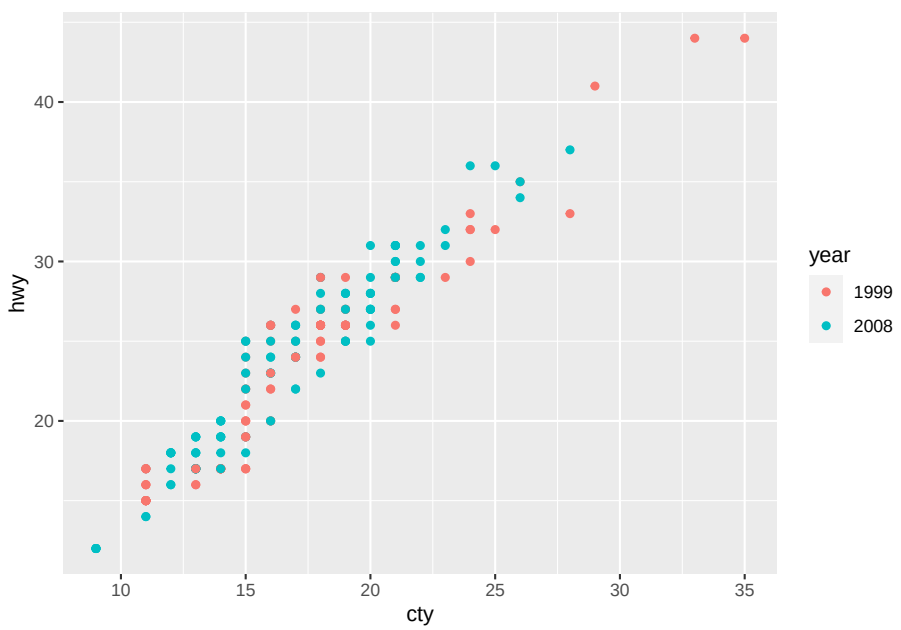
geom_point() understands the following aesthetics (required aesthetics are in bold):

- x
- y
- alpha
- colour
- fill
- group
- shape
- size
- stroke

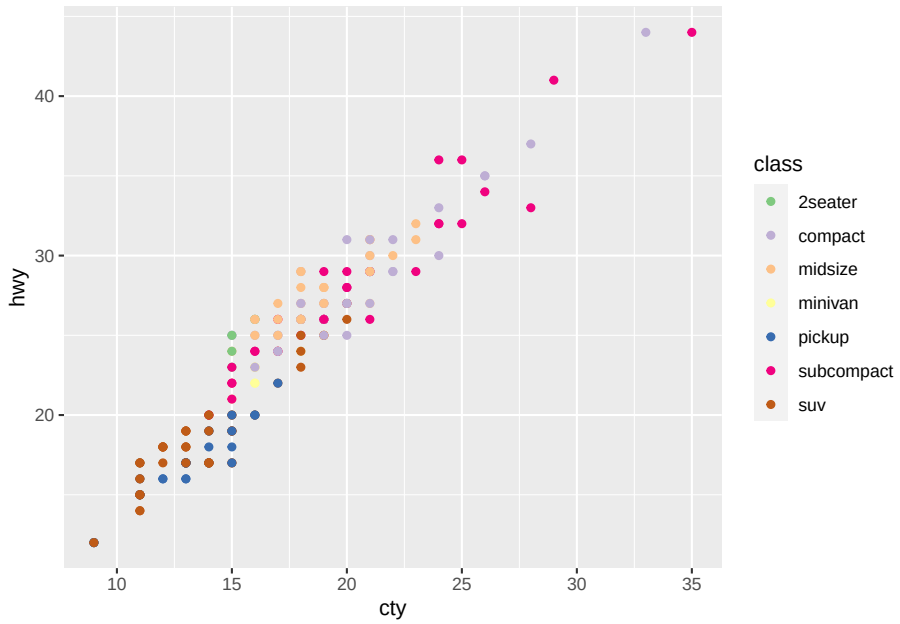
```
ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point()
```



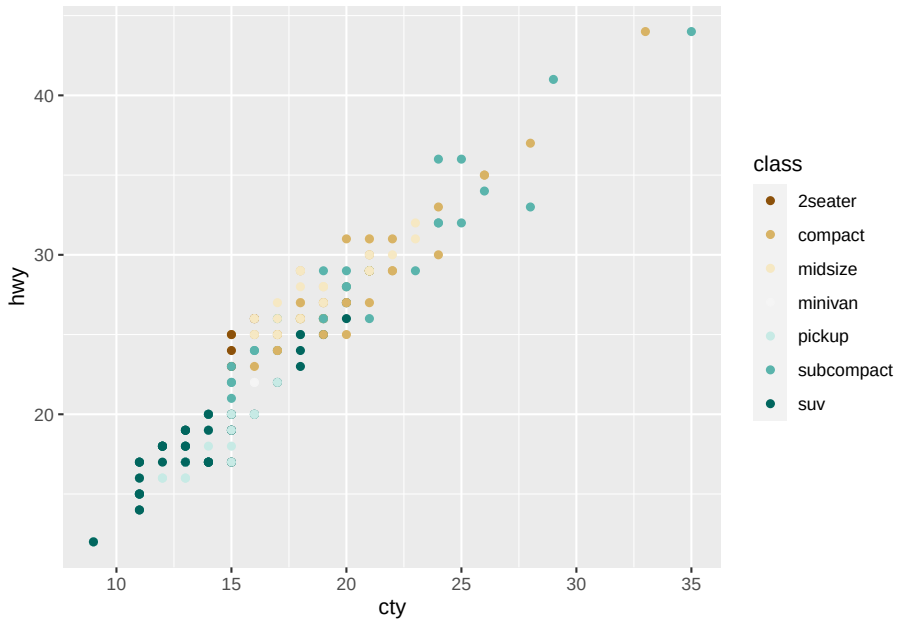

```
ggplot(dados, aes(x = cty, y = hwy, col = factor(year))) +
  geom_point() +
  labs(col = "year")
```



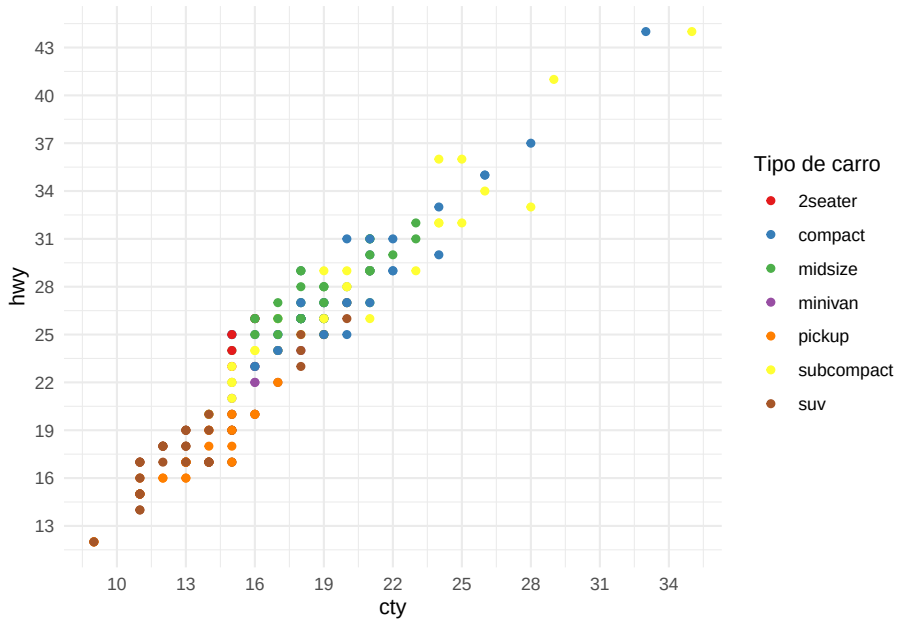
```
# Alternativa
ggplot(dados, aes(x = cty, y = hwy, col = factor(class))) +
  geom_point() +
  labs(col = "class")+
  scale_color_brewer(type = "qual")
```



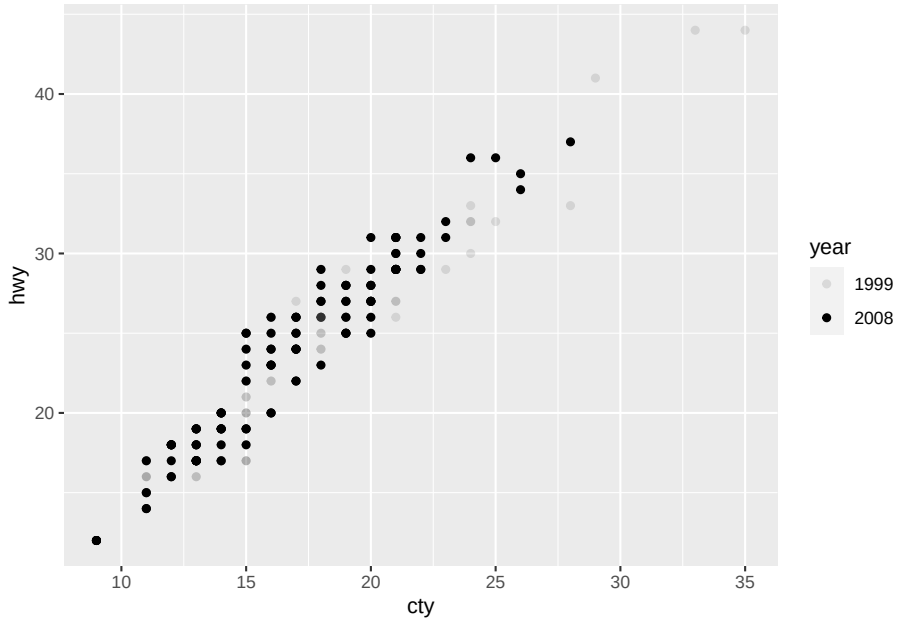

```
ggplot(dados, aes(x = cty, y = hwy, col = factor(class))) +
  geom_point() +
  labs(col = "class")+
  scale_color_brewer(type = "div")
```



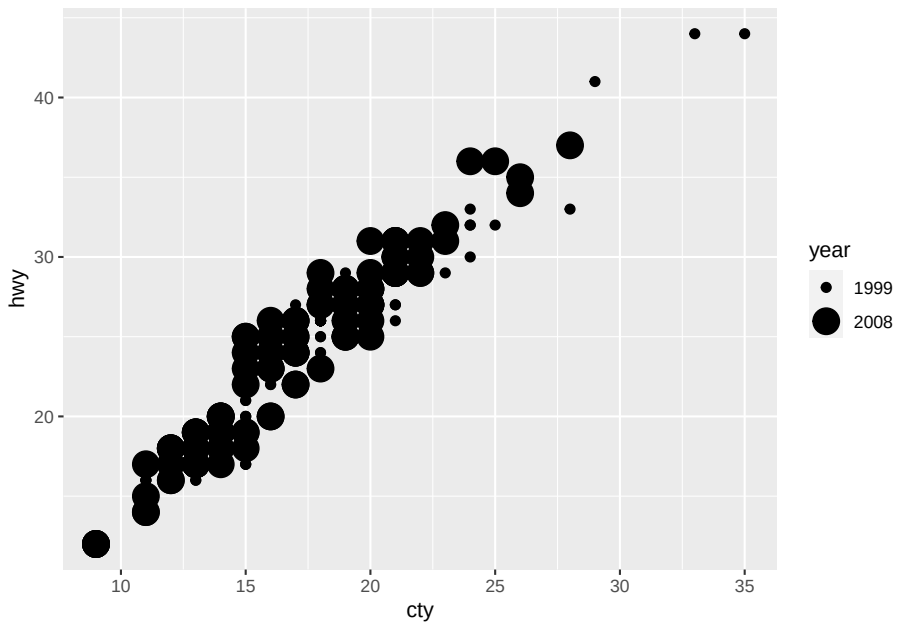

```
ggplot(dados, aes(x = cty, y = hwy, col = factor(class))) +
  geom_point() +
  labs(col = "class")+
  scale_color_brewer(palette = "Set1", name = "Tipo de carro")+
  scale_y_continuous(breaks = seq(10,60,3))+
  scale_x_continuous(breaks = seq(10,40,3))+
  theme_minimal()
```



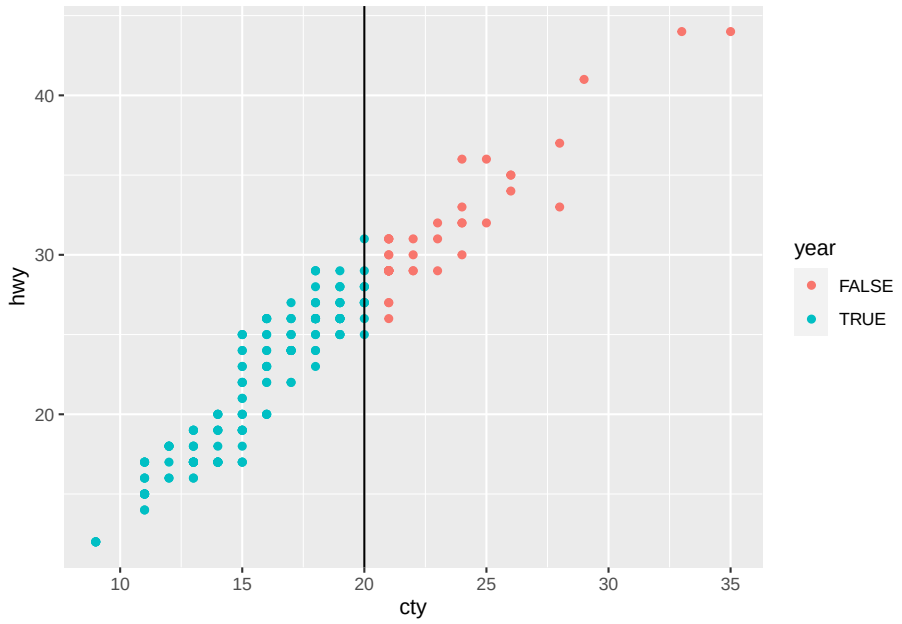

```
ggplot(dados, aes(x = cty, y = hwy, alpha = factor(year))) +
  geom_point() +
  labs(alpha = "year")
```



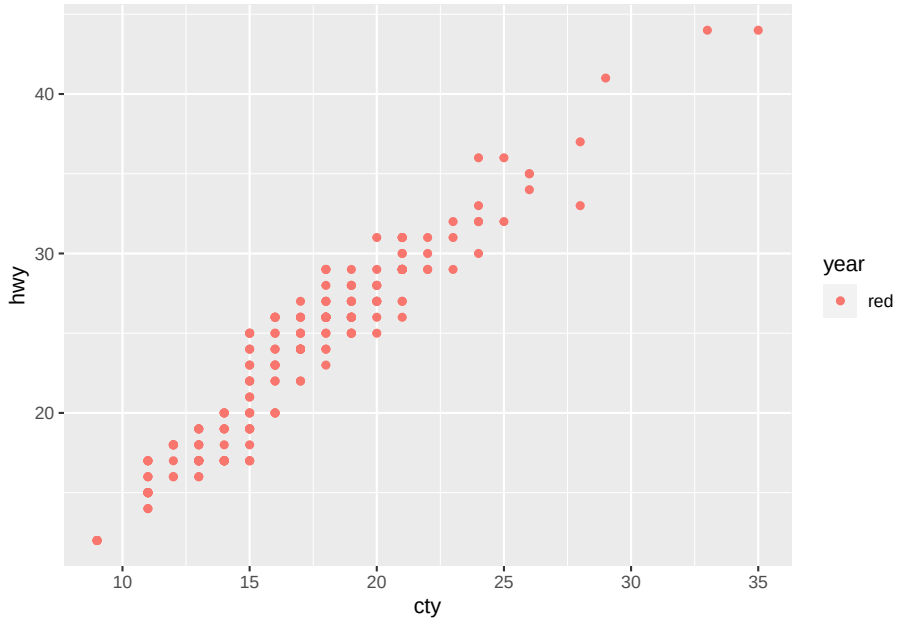
```
ggplot(dados, aes(x = cty, y = hwy, size = factor(year))) +
  geom_point() +
  labs(size = "year")
```



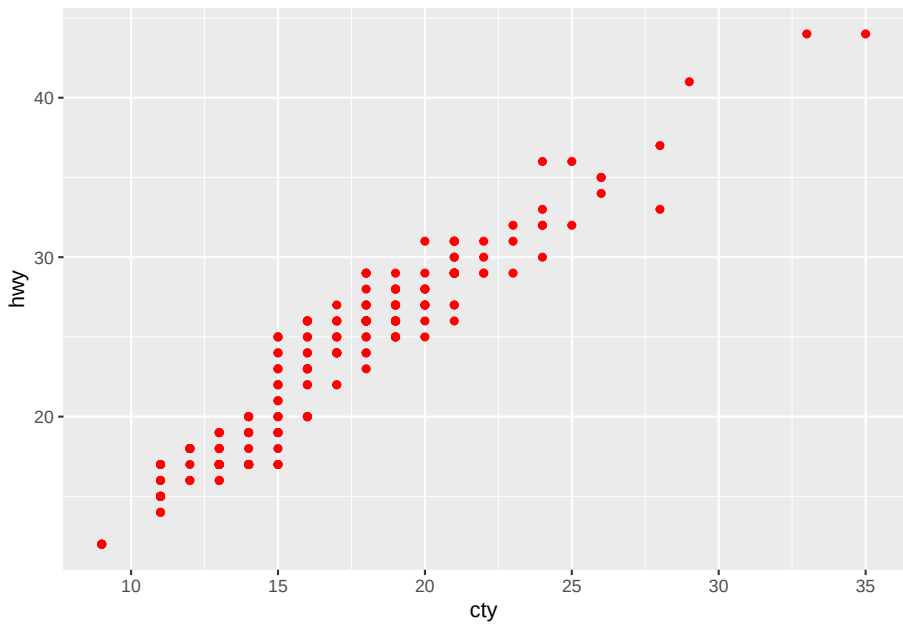

```
# Alternativa
ggplot(dados, aes(x = cty, y = hwy, col = cty <= 20)) +
  geom_point() +
  geom_vline(xintercept = 20)+
  labs(col = "year")
```



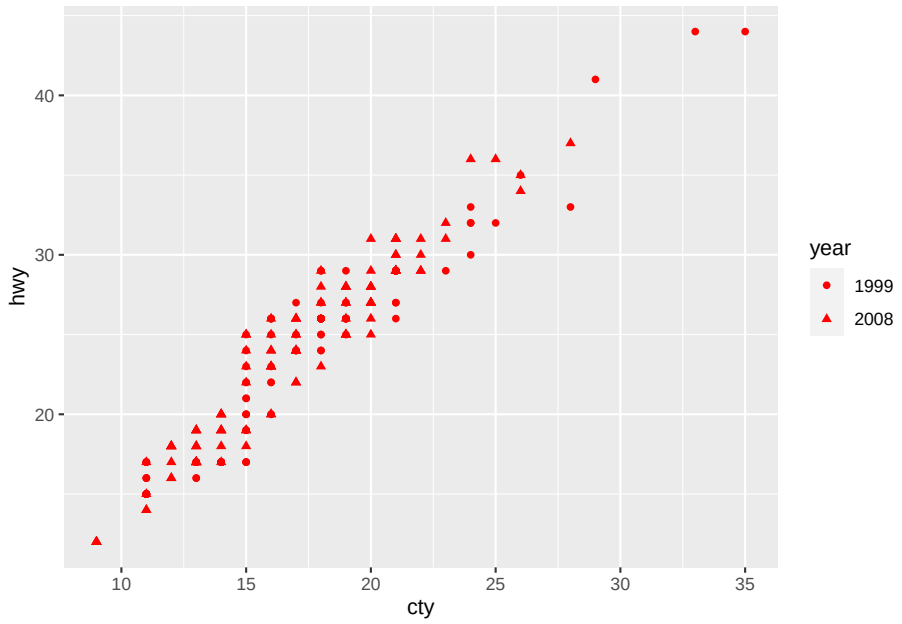

```
# Erro comum
ggplot(dados, aes(x = cty, y = hwy, col = "red")) +
  geom_point()+
  labs(col = "year")
```



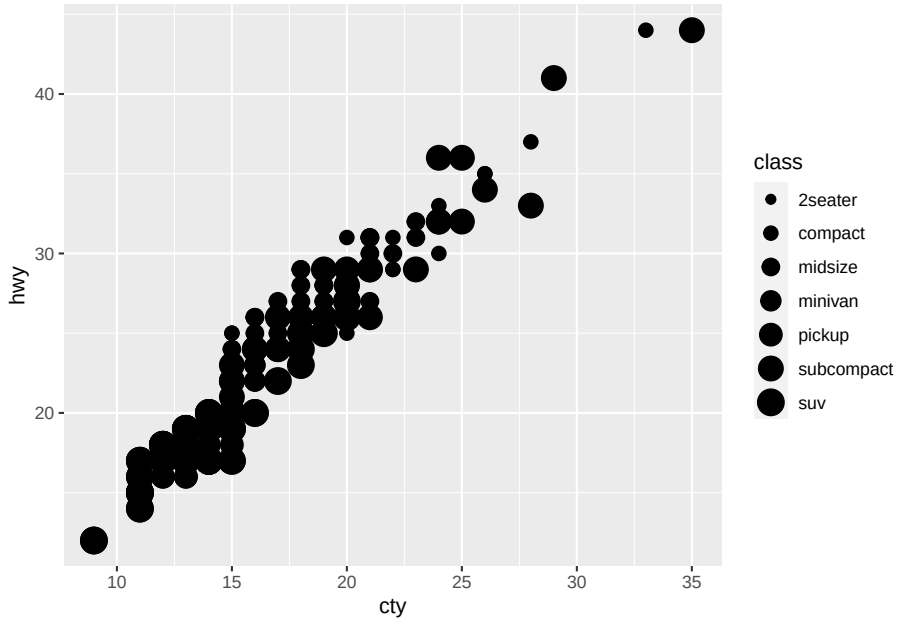
```
ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point(col = "red")+
  labs(col = "year")
```



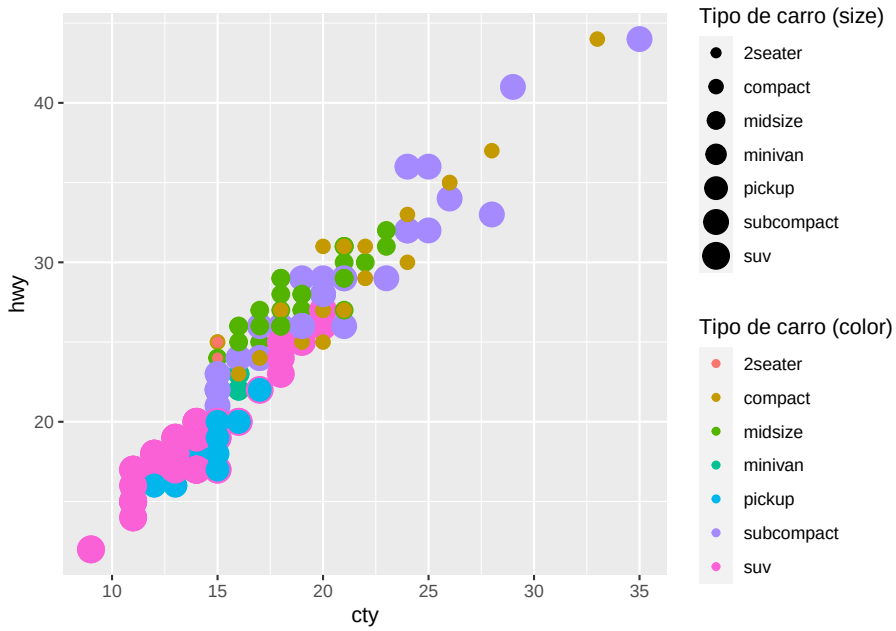

```
# Fim  Erro comum

ggplot(dados, aes(x = cty, y = hwy, shape = factor(year))) +
  geom_point(col = "red") +
  labs(shape = "year")
```



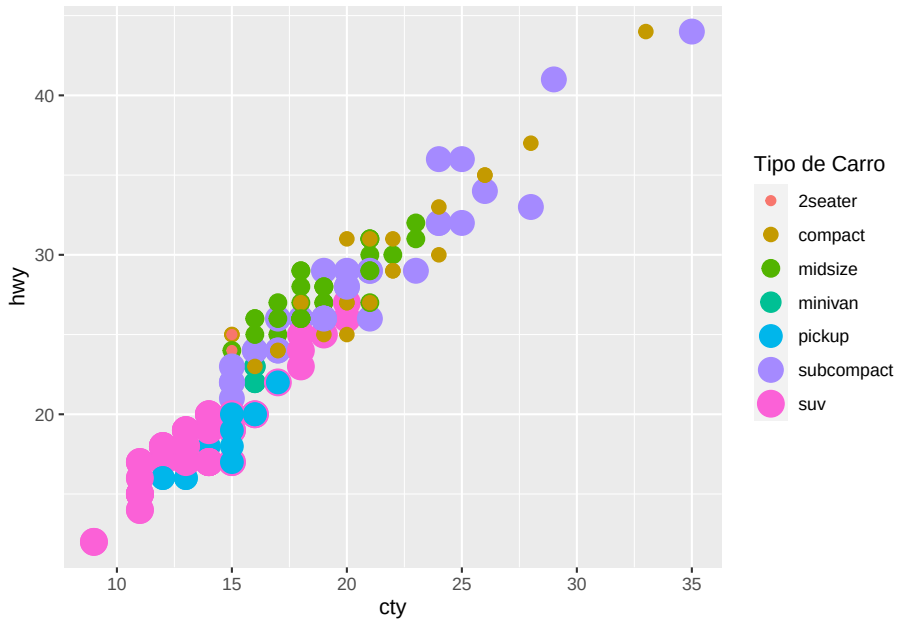

```
ggplot(dados, aes(x = cty, y = hwy, size = class)) +
  geom_point() +
  labs(size = "class")
```



```
ggplot(dados, aes(x = cty, y = hwy,
                  size = class,
                  col = class)) +
  geom_point() +
  guides(colour = guide_legend("Tipo de carro (color)"),
         size = guide_legend("Tipo de carro (size)"))
```

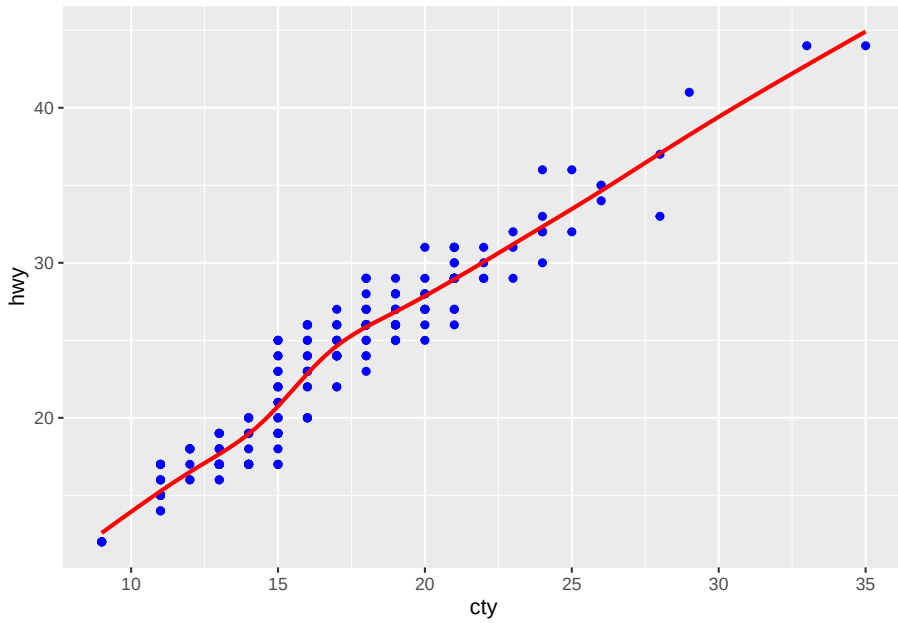



```
ggplot(dados, aes(x = cty, y = hwy,
                  size = class,
                  col = class)) +
  geom_point() +
  labs(col = "Tipo de Carro", size = "Tipo de Carro")+
  guides(col = "legend")
```

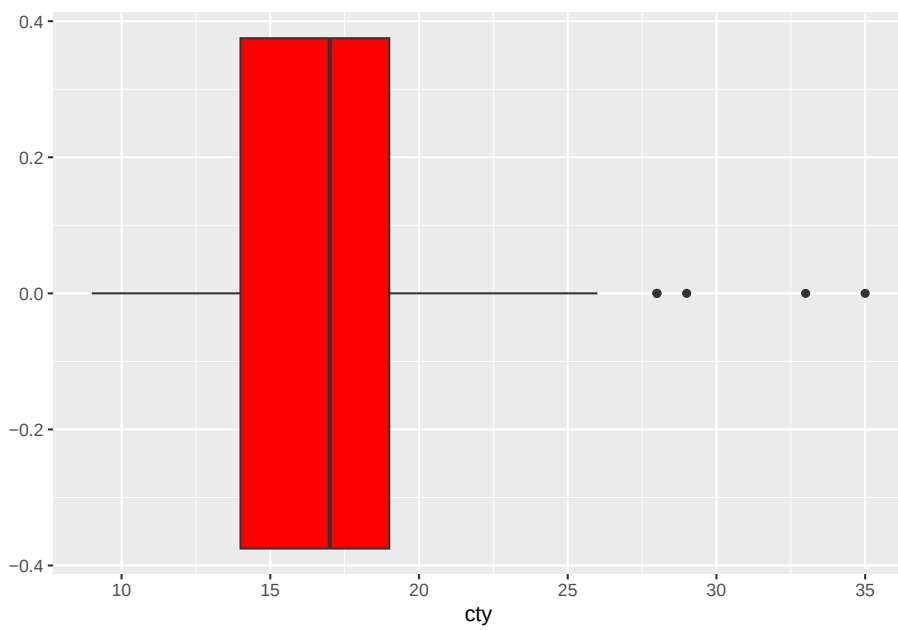


3.3 smooth, boxplot, histogram

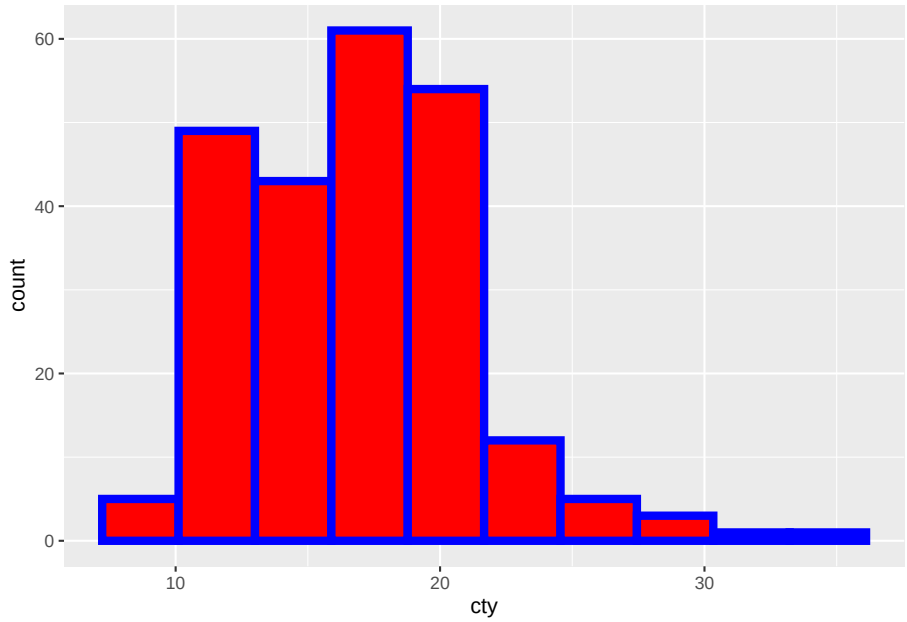
```
v1<- ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point(col = "blue")+
  geom_smooth(method = mgcv::gam,
              formula = y ~ s(x, bs = "cs") ,
              col = "red",
              se = FALSE)
v1
```



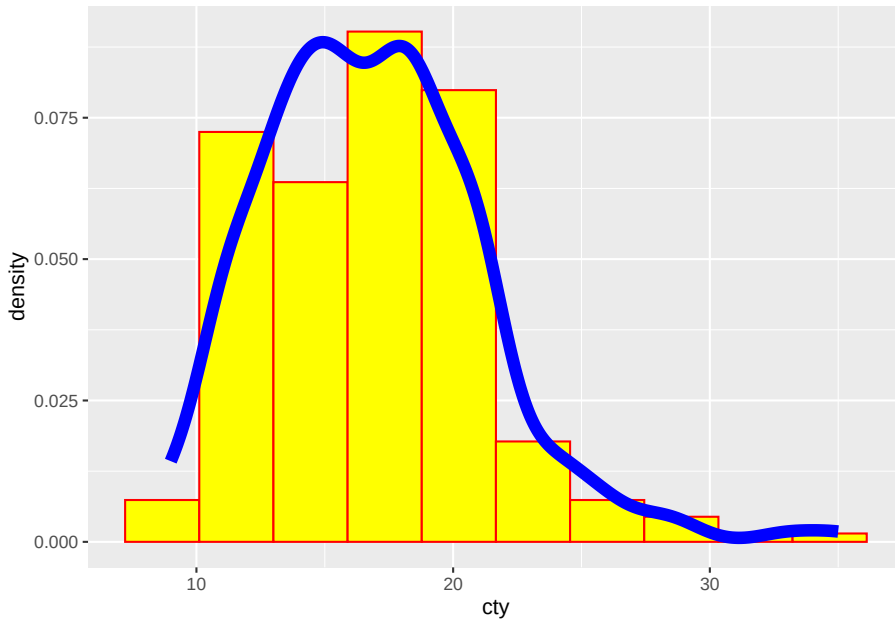

```
v2 <- ggplot(dados, aes(x = cty)) +
  geom_boxplot(fill = "red")
v2
```



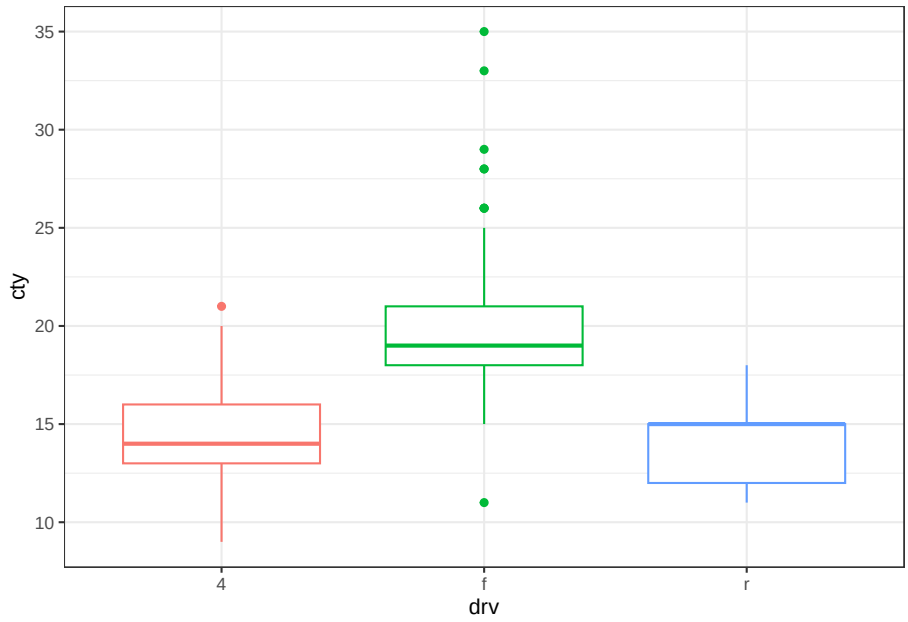
```
v3 <- ggplot(dados, aes(x = cty)) +
  geom_histogram(bins = 10, fill = "red", col = "blue", lwd=2)
v3
```




```
v4<- ggplot(dados, aes(x = cty)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 10, fill = "yellow", col = "red") +
  geom_density(col = "blue", lwd =3)
v4
```




```
# Adicional (estatístic experimental)
ggplot(dados, aes(x = drv, y = cty, col = drv)) +
  geom_boxplot()+
  theme_bw()+
  theme(legend.position = "none")
```



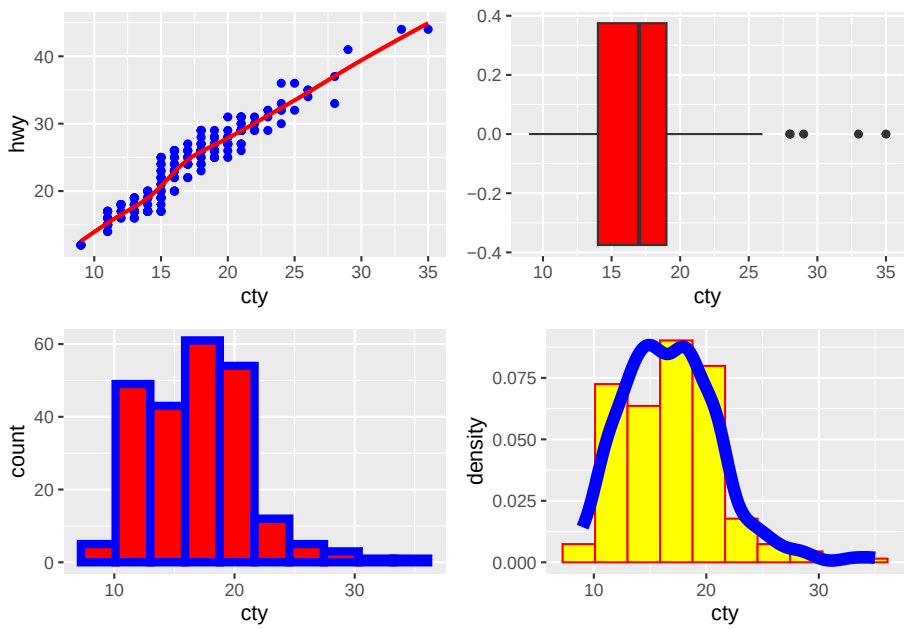
3.4 gridExtra e patchwork

Alguns links

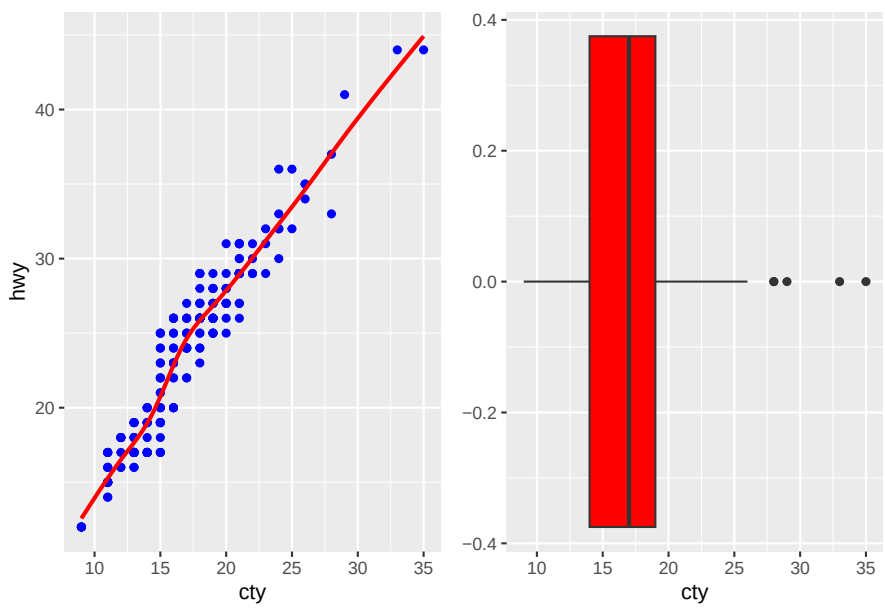
link 1: patchwork

link 2: patchwork

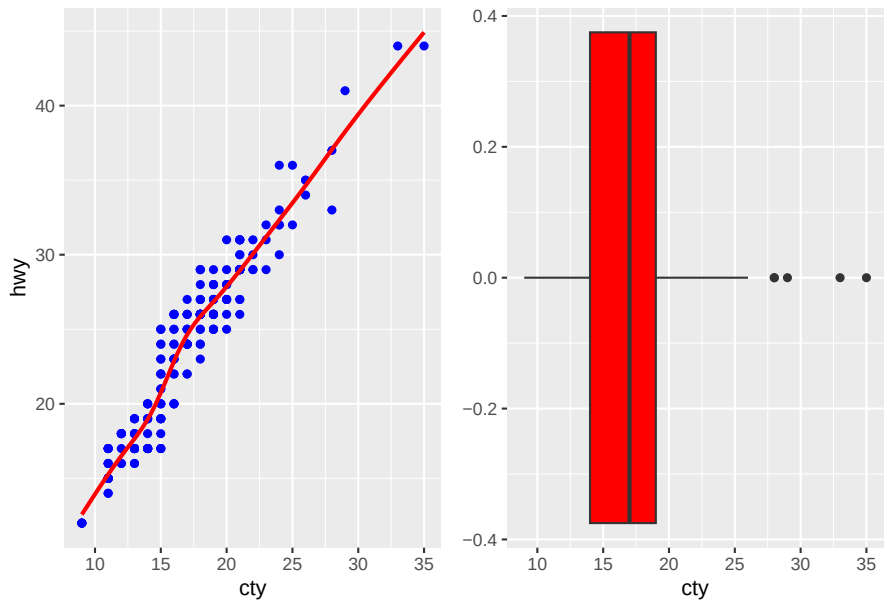
```
# gridExtra
grid.arrange(v1, v2, v3, v4)
```



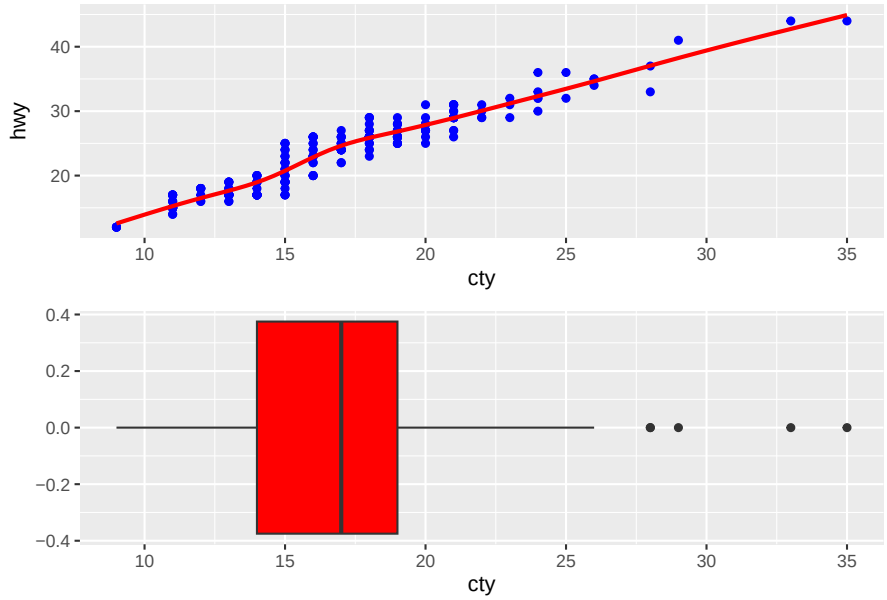

```
# patchwork
v1 + v2
```



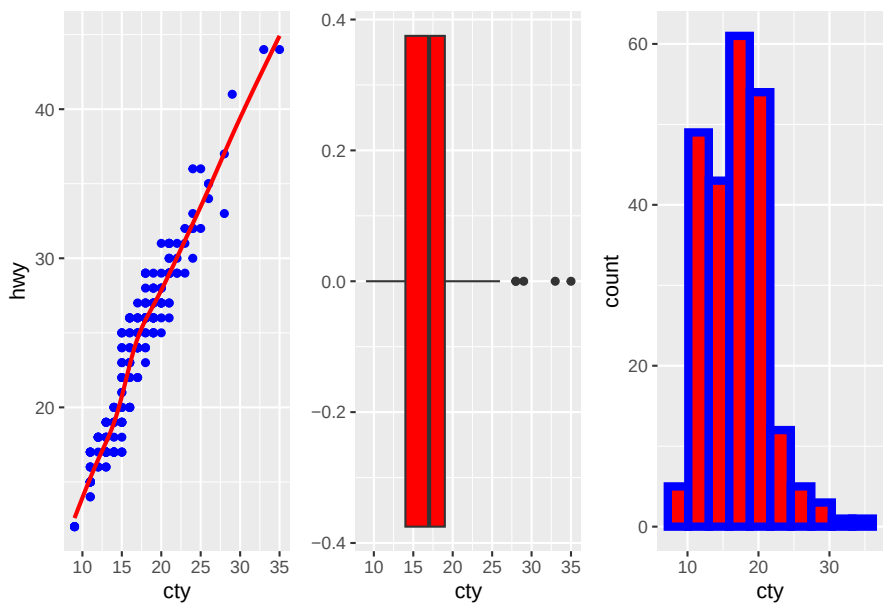
```
v1 | v2
```


hwy
40
30
20
10
15
20
25
30
35
cty
0.4
0.2
0.0
−0.2
−0.4
10
15
20
25
30
35
cty

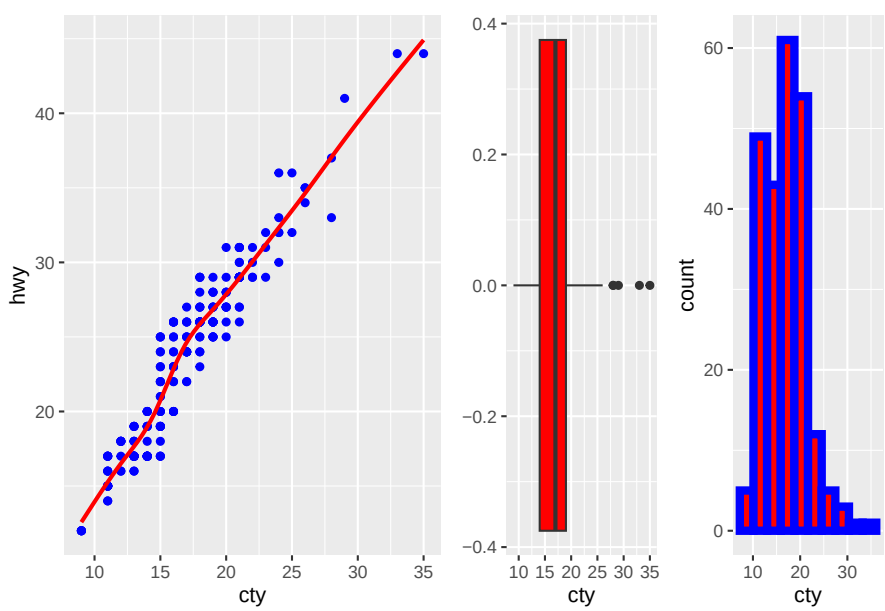
```
v1 / v2
```


hwy
40
30
20
10
15
20
25
30
35
cty
0.4
0.2
0.0
−0.2
−0.4
10
15
20
25
30
35
cty

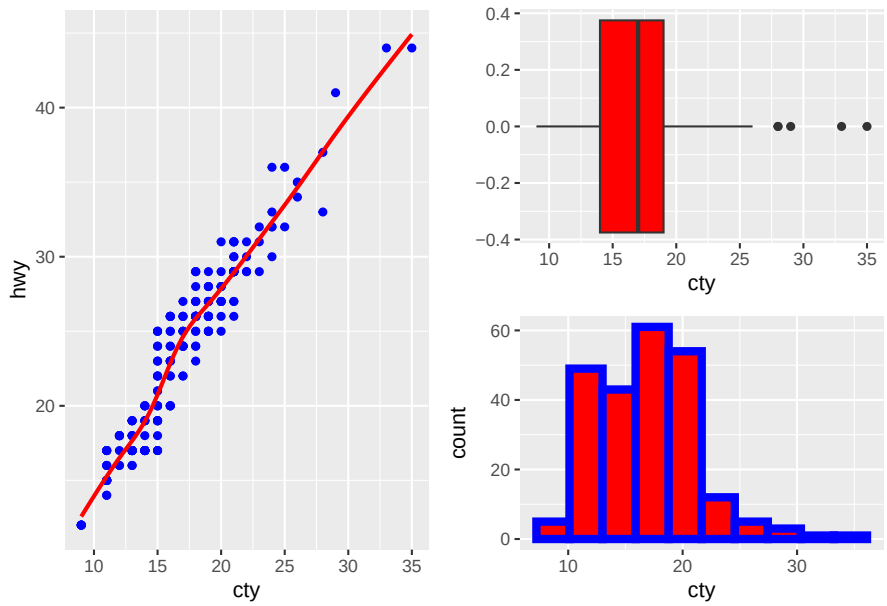
```
v1 + v2 + v3
```



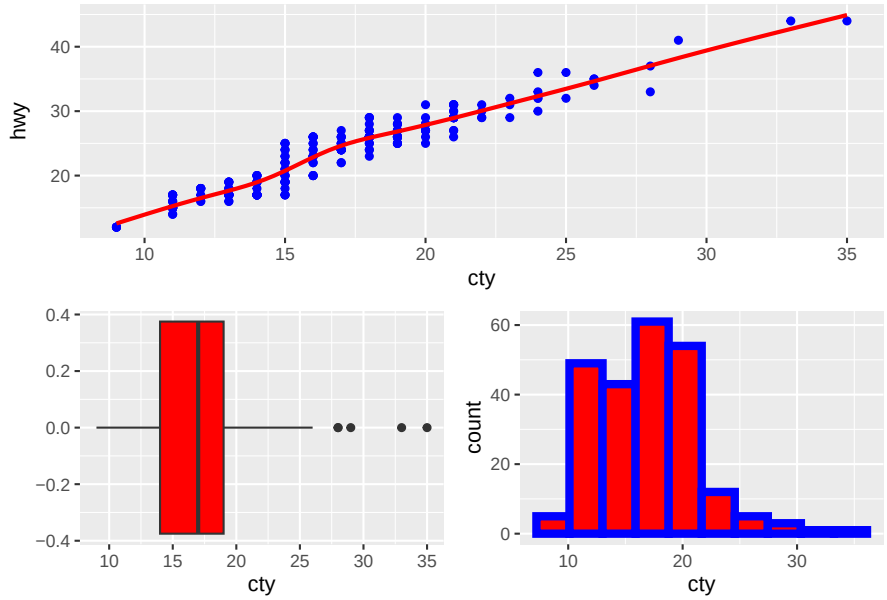

```
v1 + (v2 + v3)
```



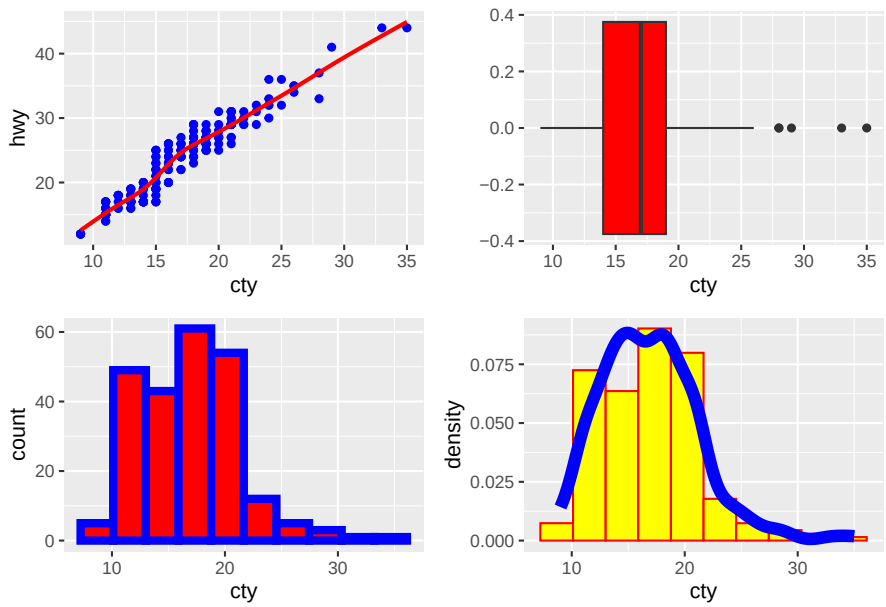
```
v1 | (v2 / v3)
```



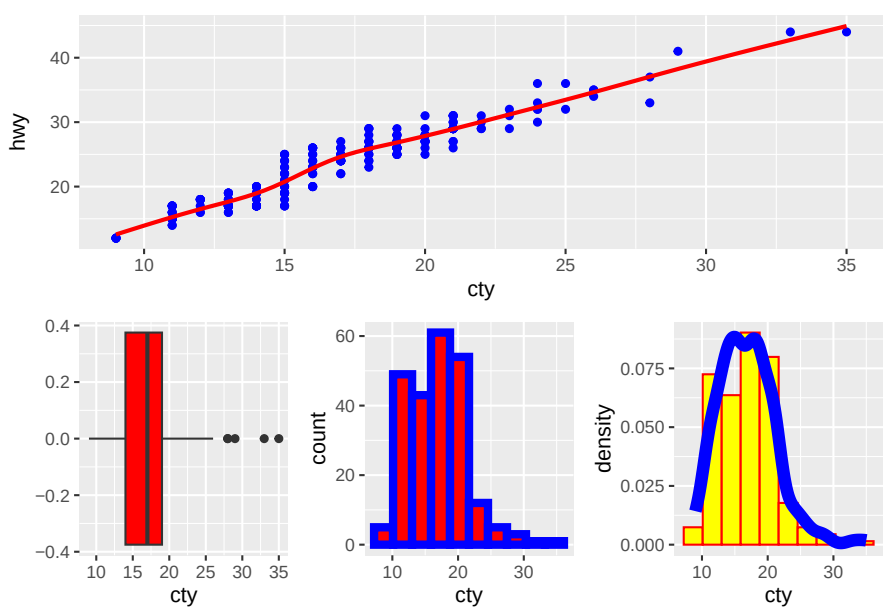

```
v1 / (v2 + v3)
```



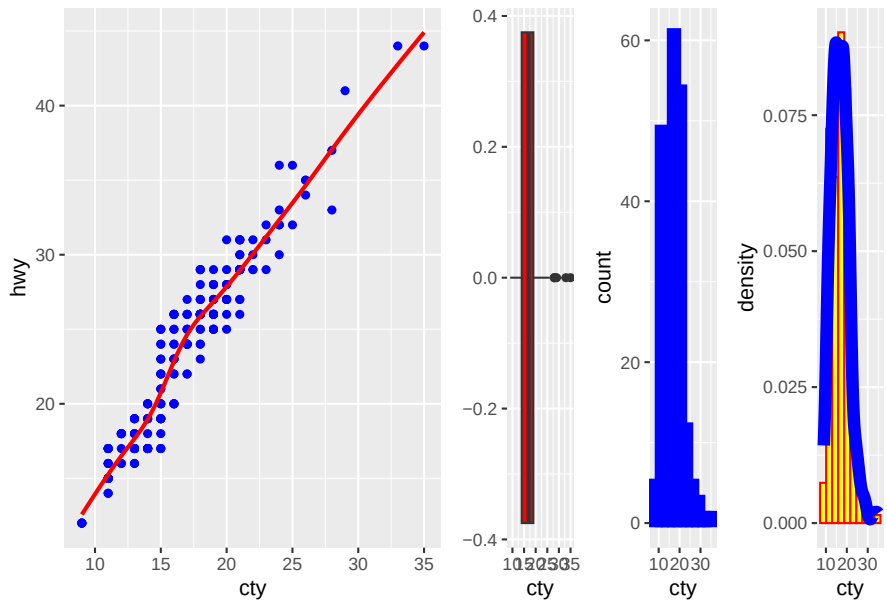
```
v1 + v2 + v3 + v4
```



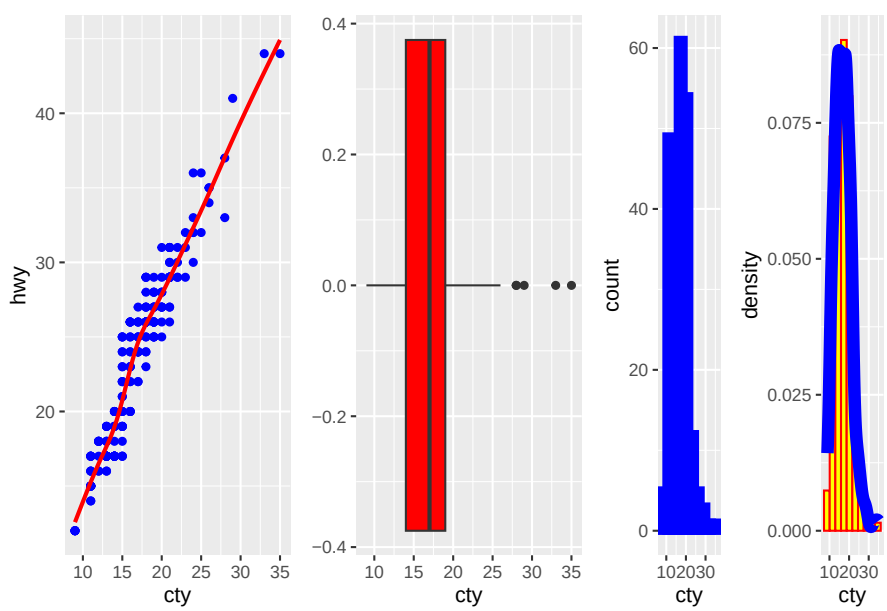

```
v1/(v2+v3+v4)
```



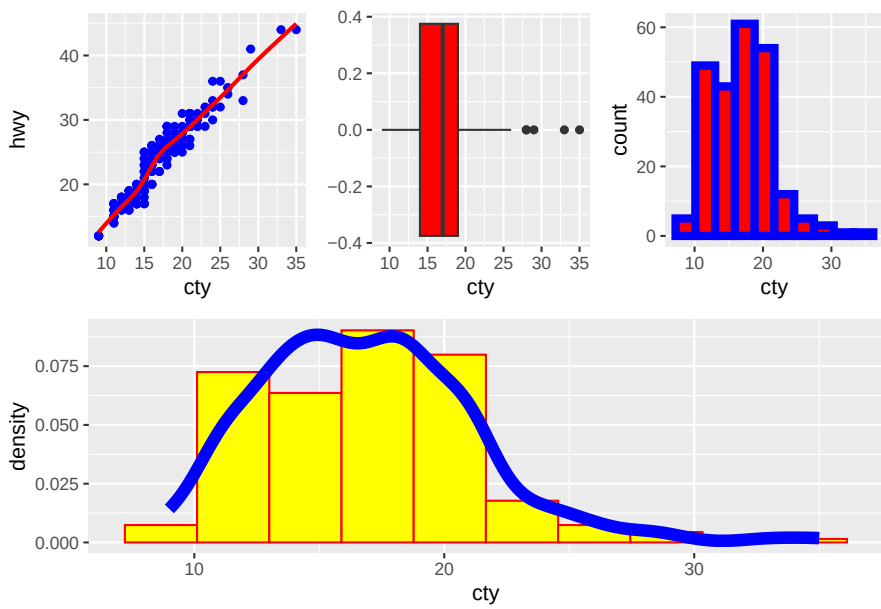
```
v1  + (v2 + v3 + v4)
```




```
v1  + v2 + (v3 + v4)
```

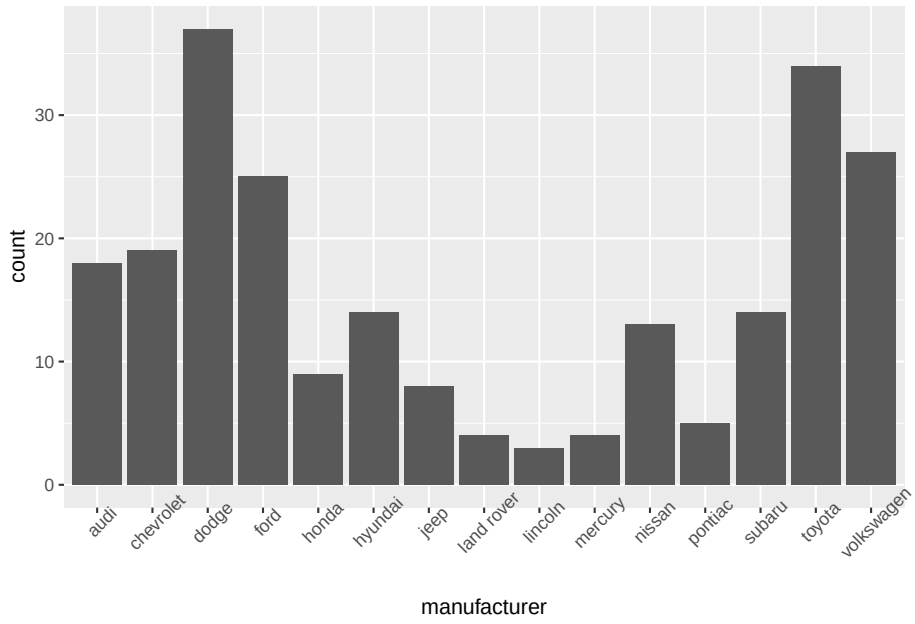


```
(v1 | v2 | v3) / v4
```

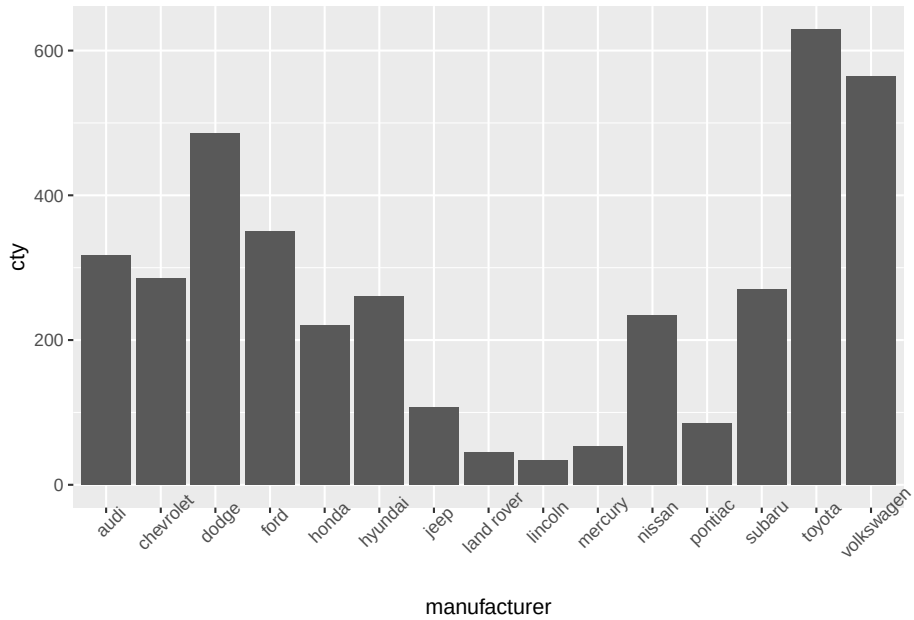


3.5 bar, col, density, density2d

```
v5 <- ggplot(dados , aes(x = manufacturer)) +
  geom_bar()+
  theme(axis.text.x = element_text(angle = 45))
v5
```



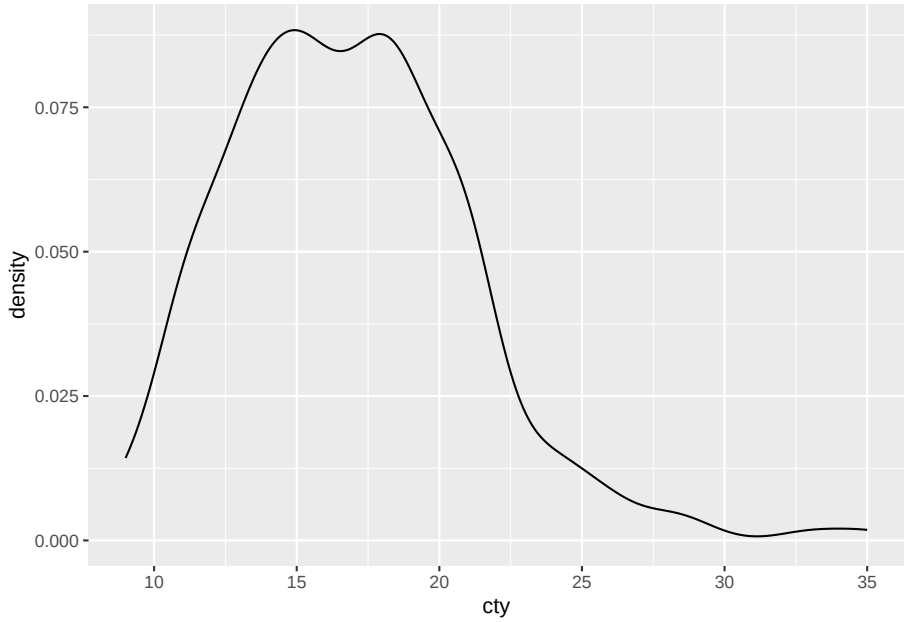

```
# Dúvidas no geom_col
v6 <- ggplot(dados , aes(x = manufacturer, y = cty)) +
  geom_col()+
  theme(axis.text.x = element_text(angle = 45))
v6
```



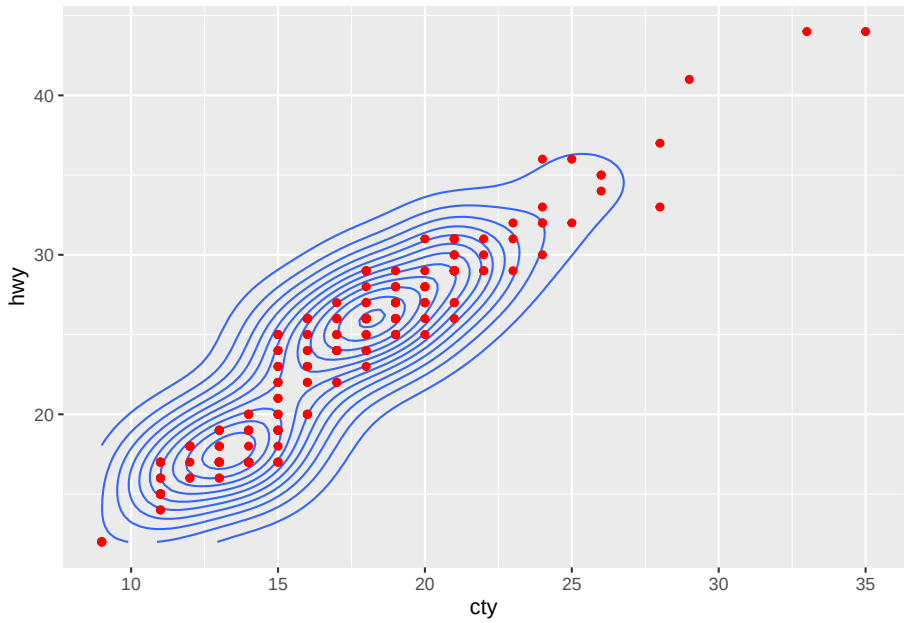

```
dados %>%
  select(manufacturer, cty) %>%
  group_by(manufacturer) %>%
  summarise(soma_total_cty = sum(cty),
            n = n())
```

```
## # A tibble: 15 x 3
##    manufacturer soma_total_cty     n
##    <fct>                 <int> <int>
##  1 audi                    317    18
##  2 chevrolet               285    19
##  3 dodge                   486    37
##  4 ford                    350    25
##  5 honda                   220     9
##  6 hyundai                 261    14
##  7 jeep                    108     8
##  8 land rover               46     4
##  9 lincoln                  34     3
## 10 mercury                  53     4
## 11 nissan                  235    13
## 12 pontiac                  85     5
## 13 subaru                  270    14
## 14 toyota                  630    34
## 15 volkswagen              565    27
```

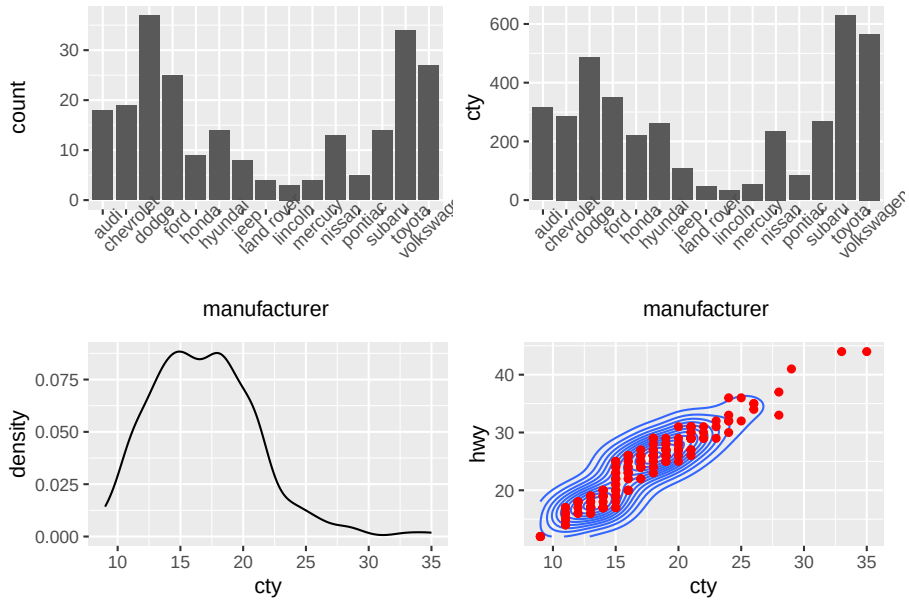
```
# dados %>%
#   filter(manufacturer == "audi") %>%
#   select(cty) %>%
#   sum()
v7 <- ggplot(dados , aes(x = cty)) +
  geom_density()
v7
```




```
v8 <- ggplot(dados, aes(x = cty, y = hwy)) +
  geom_density2d()+
  geom_point(colour = "red")
v8
```


hwy
cty
40
30
20
10
15
20
25
30
35

```
(v5+v6)/ (v7 + v8)
```


count
manufacturer
cty
manufacturer
density
cty
hwy
cty
audi
chevrolet
dodge
ford
honda
hyundai
jeep
land rover
lincoln
mercury
nissan
pontiac
subaru
toyota
volkswagen

```r
# Deixar pra depois...
 dados %>%
    select(manufacturer, hwy, year) %>%
    filter(manufacturer == "audi", year == "1999") %>%
    summarise(media = max(hwy))
```

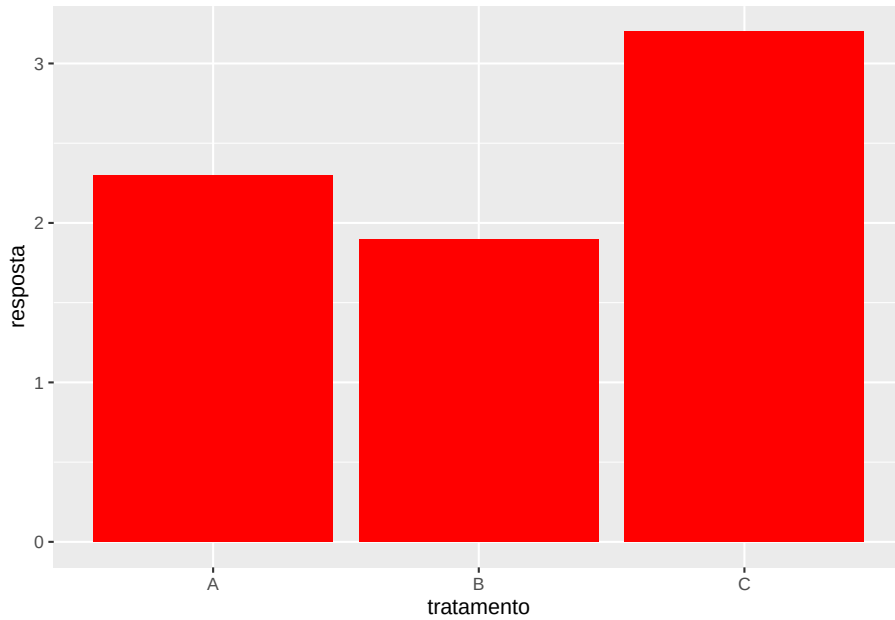
```
## # A tibble: 1 x 1
##   media
##   <int>
## 1    29
```

```r
ggplotly(
ggplot(dados, aes(x = manufacturer, y = hwy, fill = factor(year))) +
  geom_col(position = "dodge") +
  labs(fill = "year") +
  theme(axis.text.x = element_text(angle = 45)))

dados %>% select(manufacturer, hwy, year) %>%
  group_by(manufacturer, year) %>%
  summarise(media = mean(hwy))
```

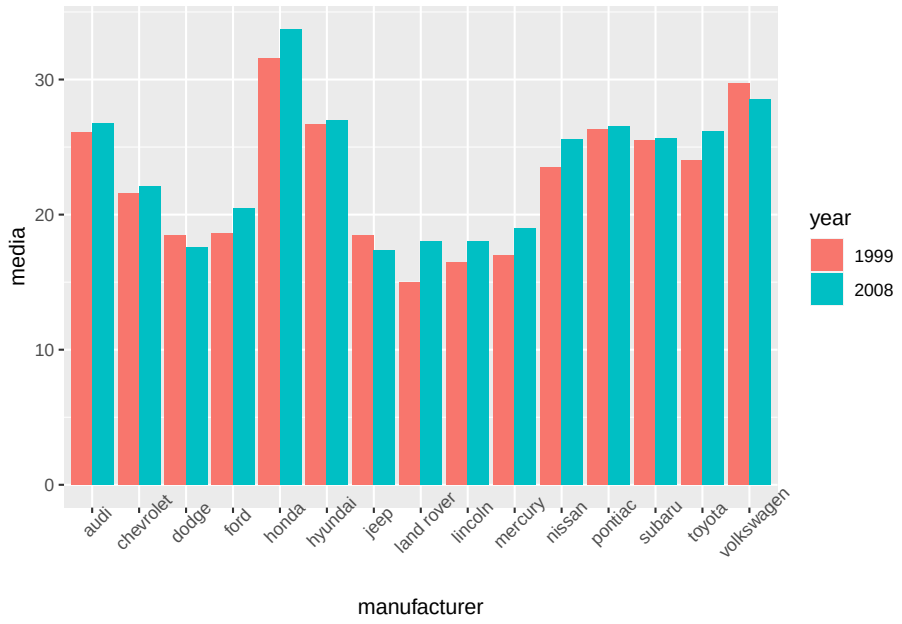
```r
# Para pensar

(dados_trat <- data.frame(tratamento = LETTERS[1:3],
                          resposta = c(2.3, 1.9, 3.2)))
```

```
##   tratamento resposta
## 1          A      2.3
## 2          B      1.9
## 3          C      3.2
```

```r
ggplot(dados_trat, aes(tratamento, resposta)) +
  geom_col(fill = "red")
```

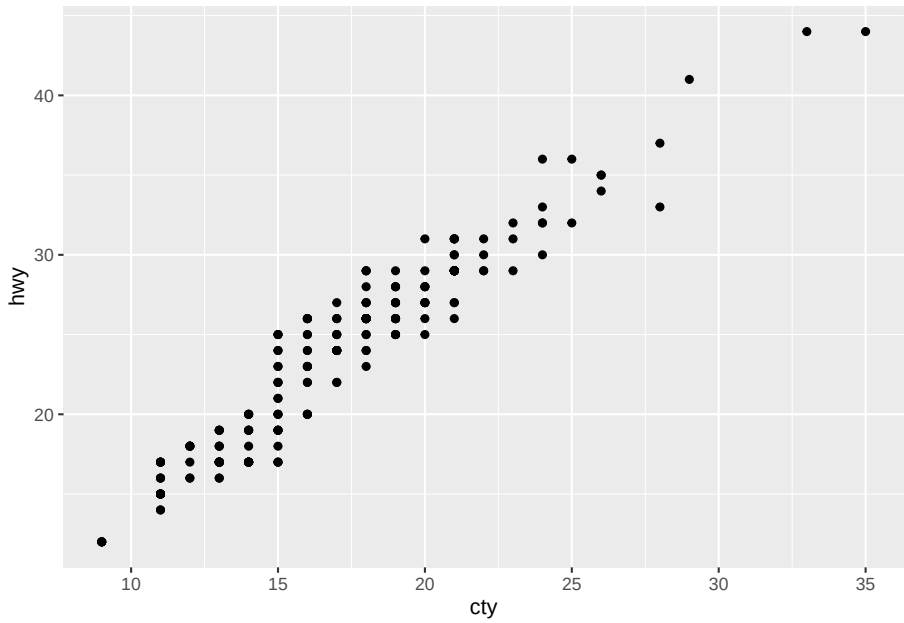



```
# Mais detalhes...
dados %>% select(manufacturer, hwy, year) %>%
  group_by(manufacturer, year) %>%
  summarise(media = mean(hwy), .groups = "drop") %>%
  ggplot(aes(x = manufacturer, y = media, fill = factor(year)))+
  geom_col(position = "dodge")+
  labs(fill = "year") +
  theme(axis.text.x = element_text(angle = 45))
```

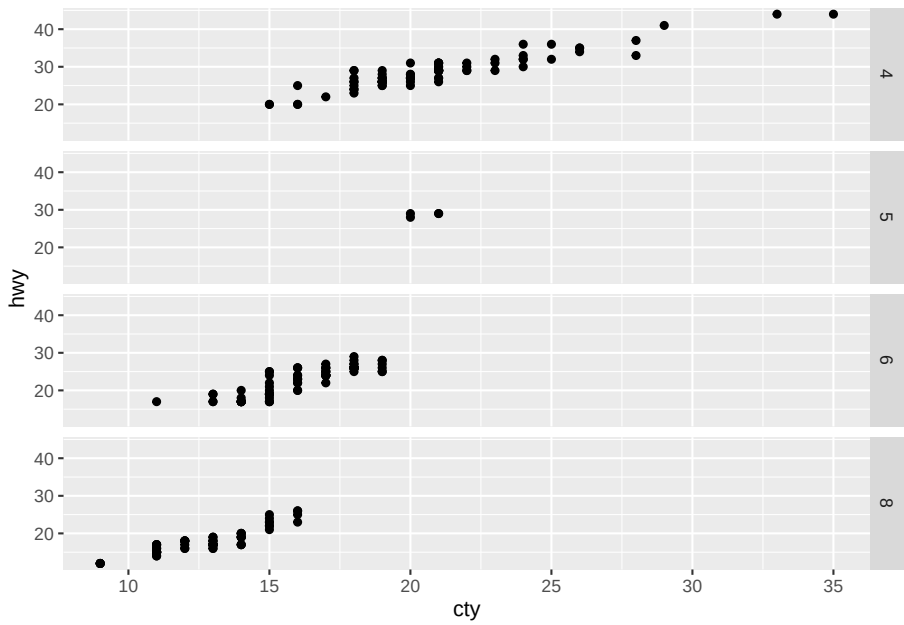


3.6 facet_grid, facet_grid

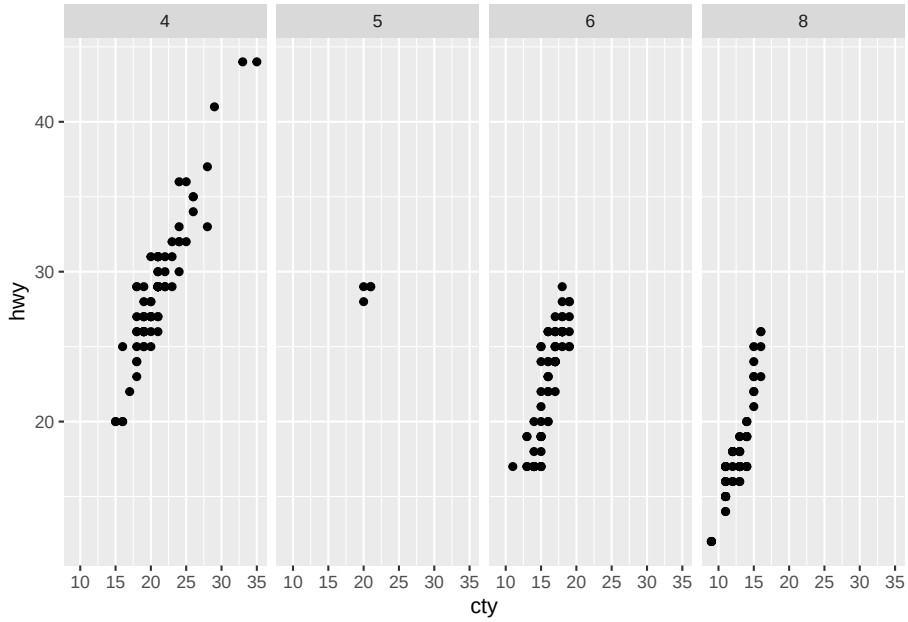
```
p1<- ggplot(dados, aes(x = cty, y = hwy)) +
  geom_point()
p1
```



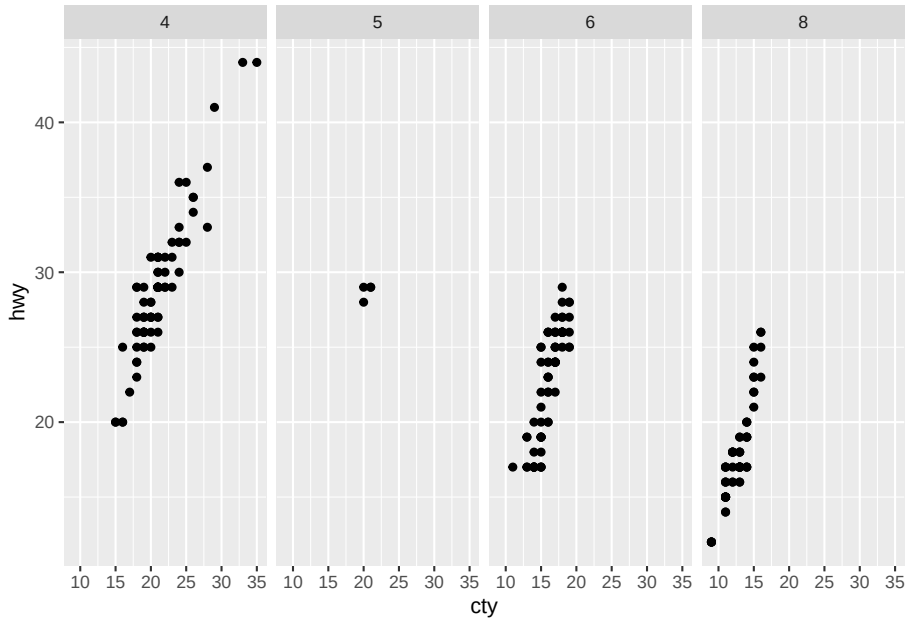

```
p1 + facet_grid(rows = vars(cyl))
```



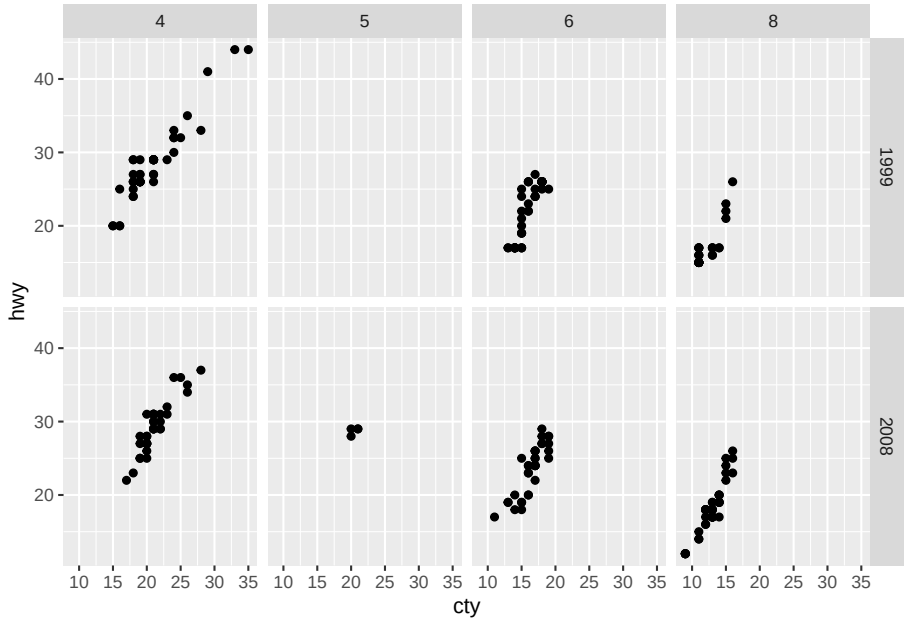
```
p1 + facet_grid(cols = vars(cyl))
```



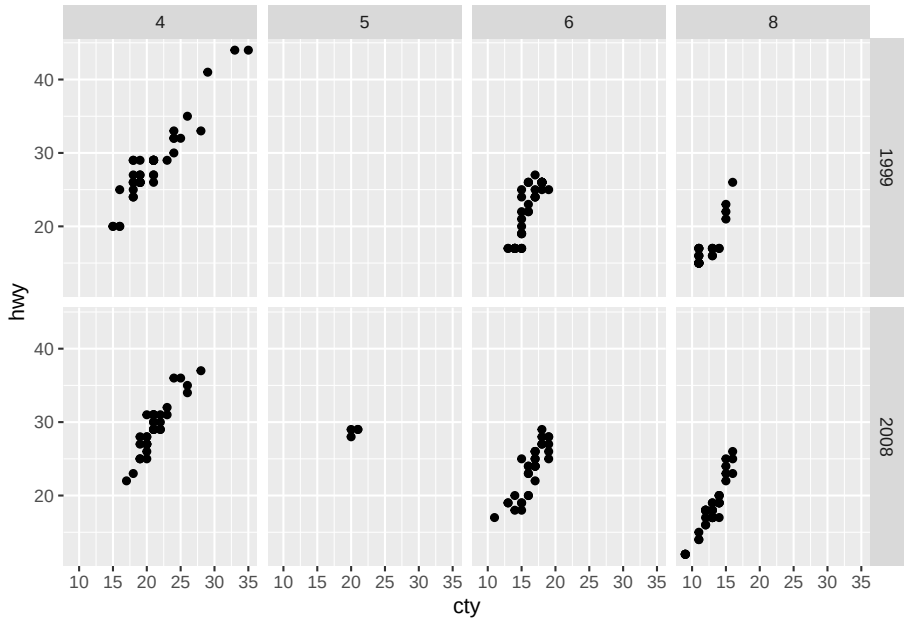

```
p1 + facet_grid(~cyl)
```



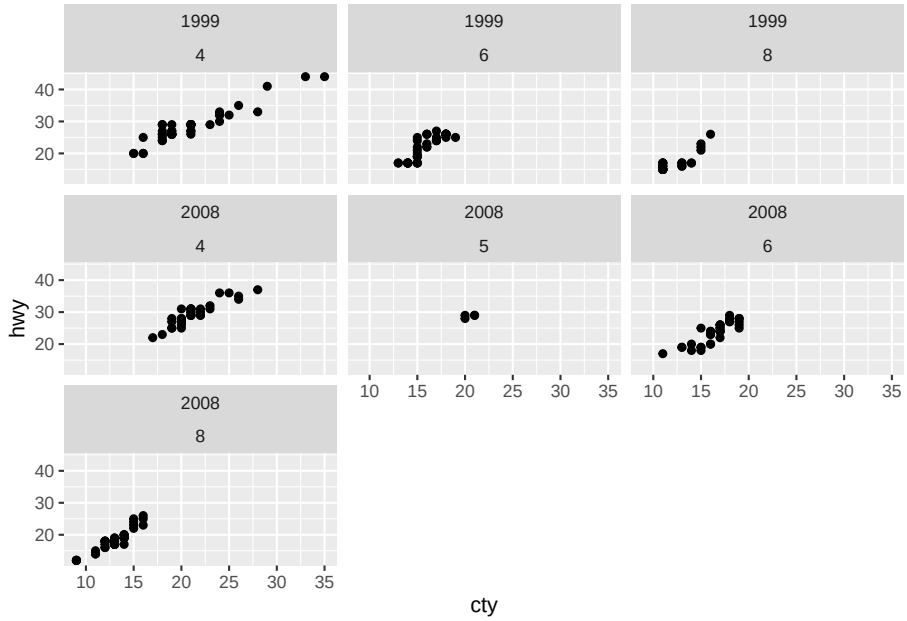
```
p1 + facet_grid(rows = vars(year), cols =vars(cyl))
```



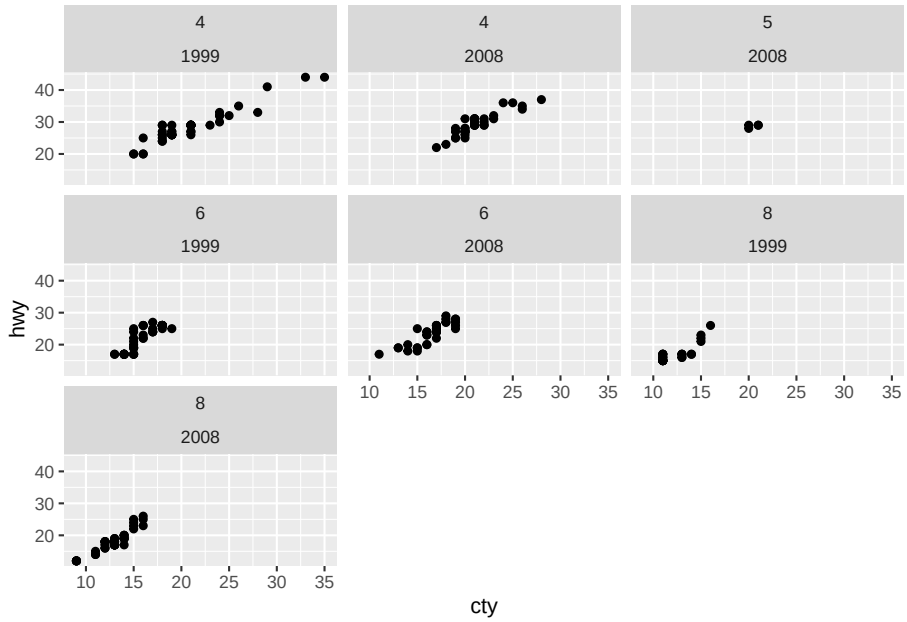

```
p1 + facet_grid(year~cyl)
```



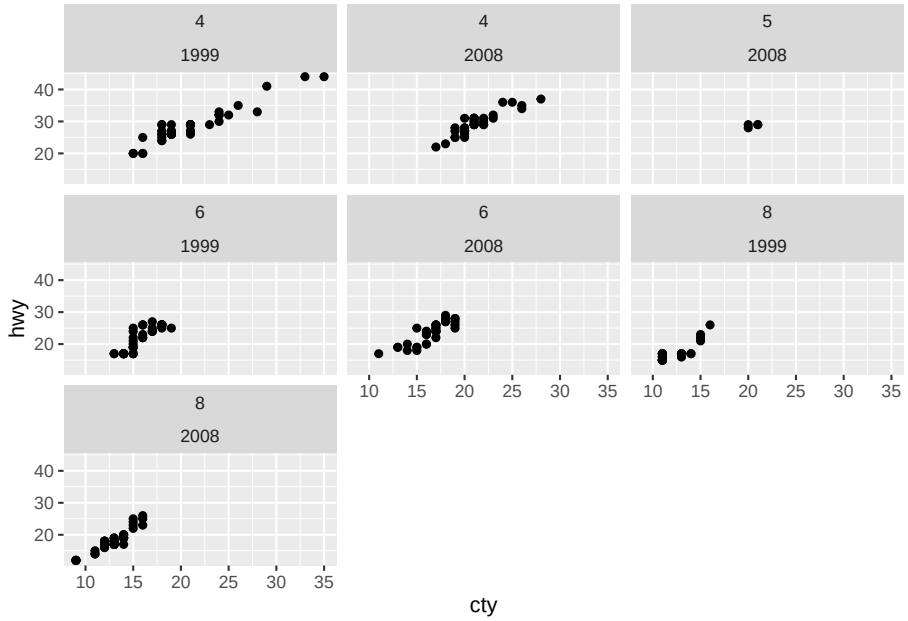
```
p1 + facet_wrap(year ~ cyl)
```



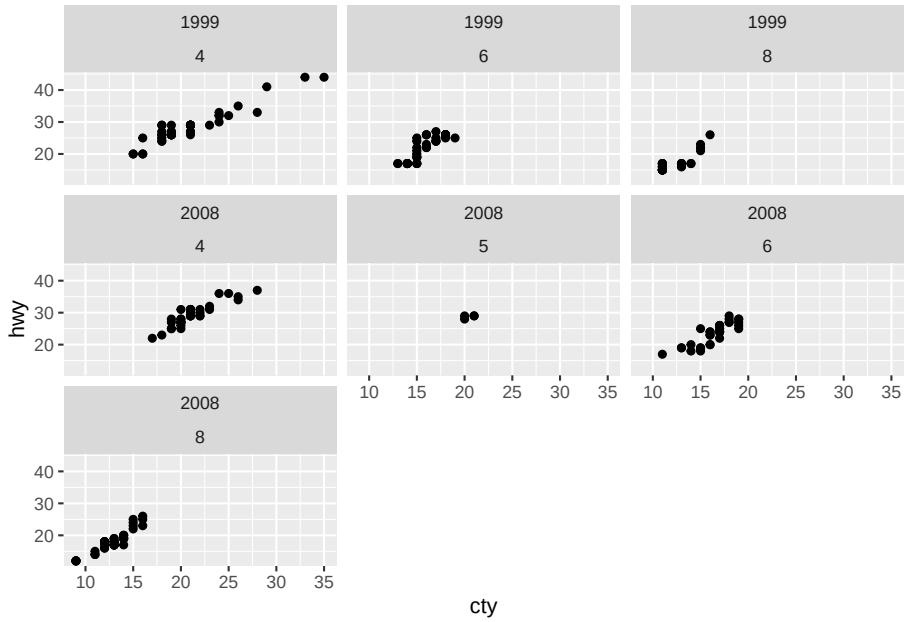

```
p1 + facet_wrap(cyl ~ year)
```



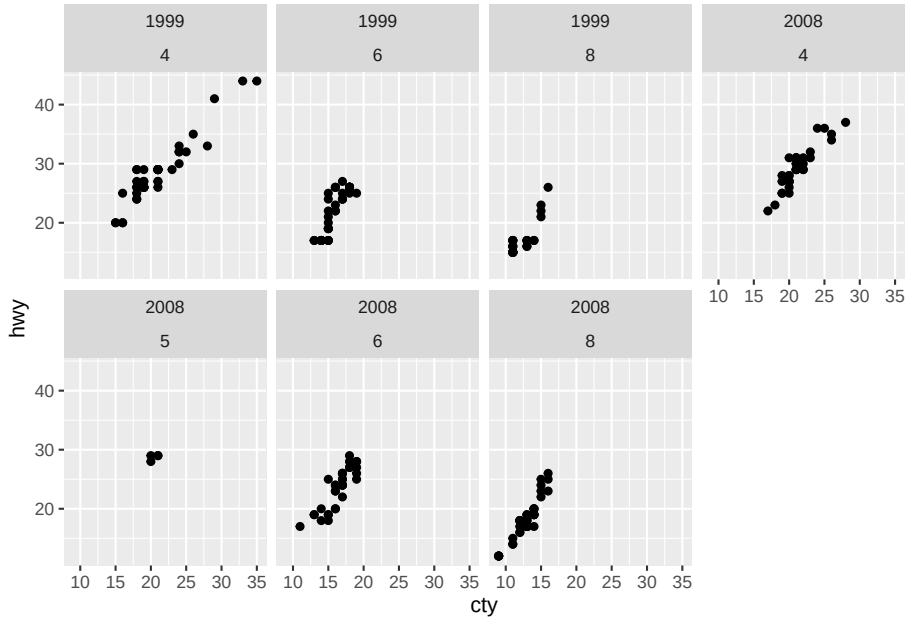
```
p1 + facet_wrap(~cyl + year)
```



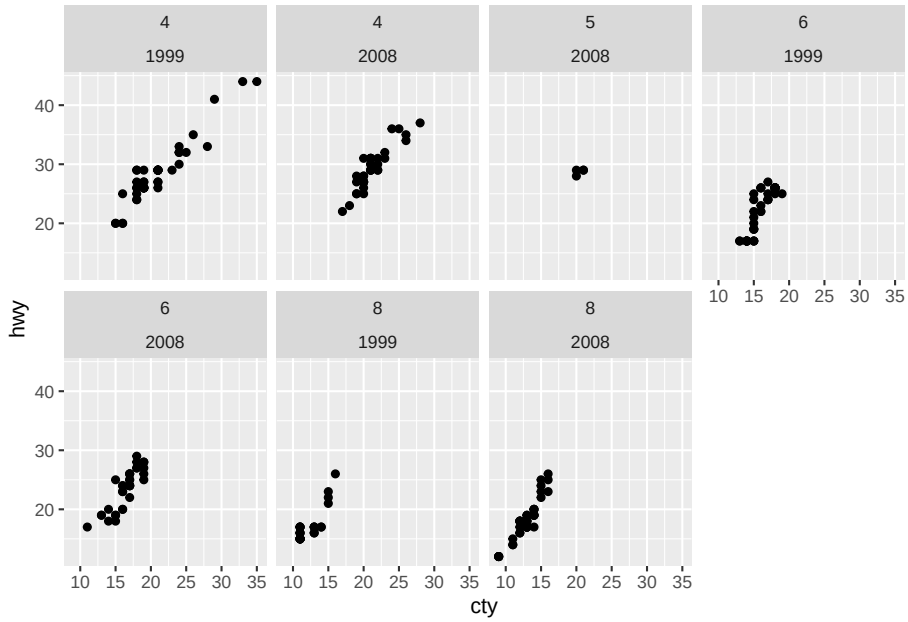

```
p1 + facet_wrap(~year + cyl)
```



```
p1 + facet_wrap(year ~ cyl, ncol = 4)
```

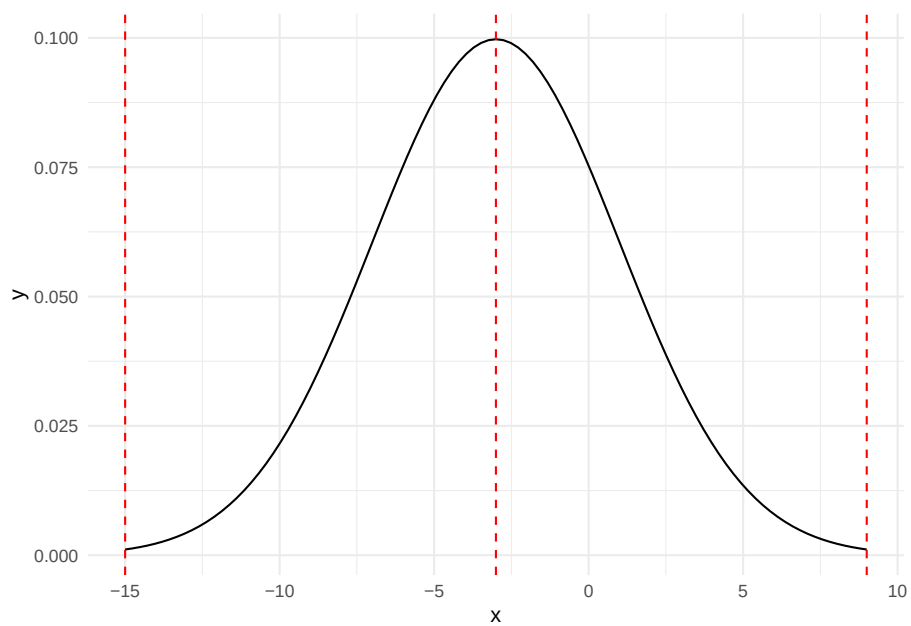



```
p1 + facet_wrap(cyl ~ year, ncol = 4)
```



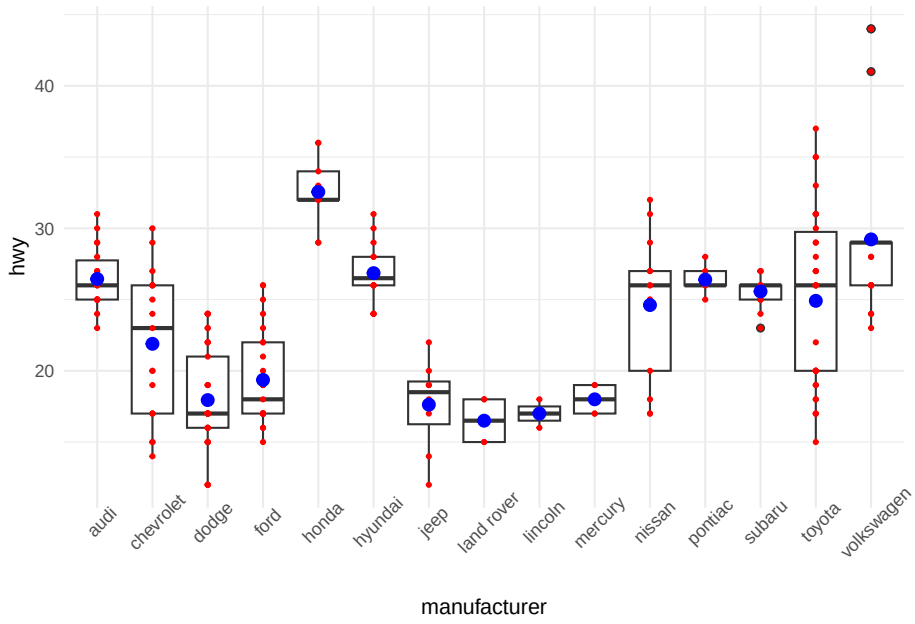
3.7 stat_function

```
a<- -3 # média
b<- 4  # desv. padrão
ggplot(data.frame(x = c(a - 3*b, a + 3*b)), aes(x)) +
  stat_function(fun = dnorm, args = list(mean = a, sd = b))+
  geom_vline(xintercept = c(a - 3*b, a, a + 3*b), col = "red", lty = 2)+
  theme_minimal()
```



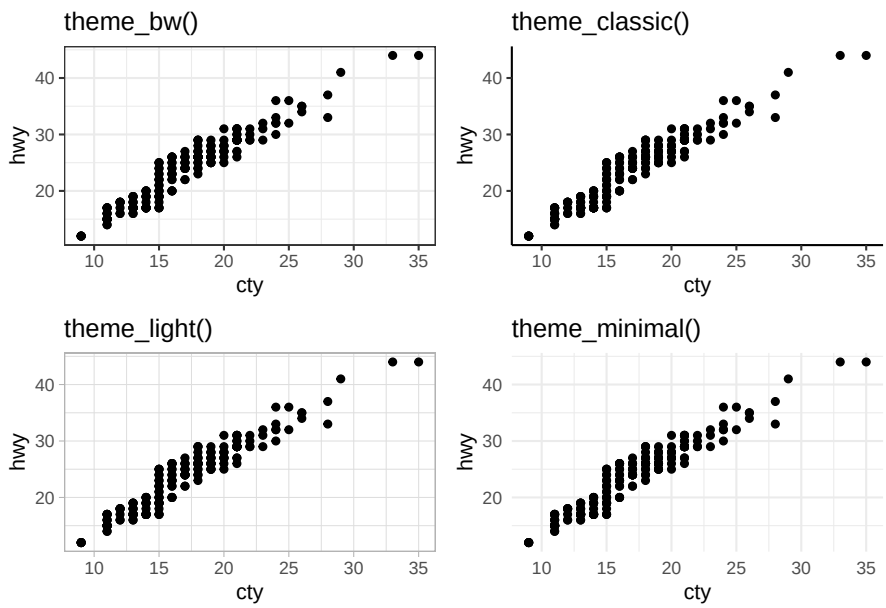
3.8 stat_summary

```
ggplot(dados, aes(x = manufacturer, y = hwy)) +
  geom_boxplot()+
  geom_point(col = "red", size=0.8)+
  stat_summary(fun = mean, col = "blue")+
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45))
```



3.9 theme_*()

```
a1<- p1 + theme_bw() + labs(title = "theme_bw()")
a2<- p1 + theme_classic() + labs(title = "theme_classic()")
a3<- p1 + theme_light() + labs(title = "theme_light()")
a4<- p1 + theme_minimal() + labs(title = "theme_minimal()")

a1 + a2 + a3 + a4
```

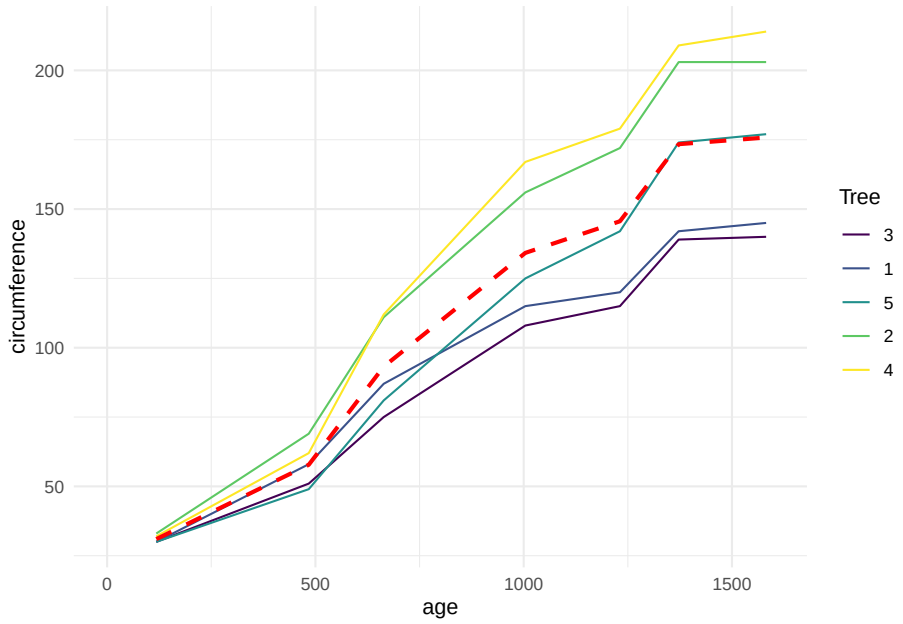


3.10 Gráfico de perfis (Spaguetti plot)

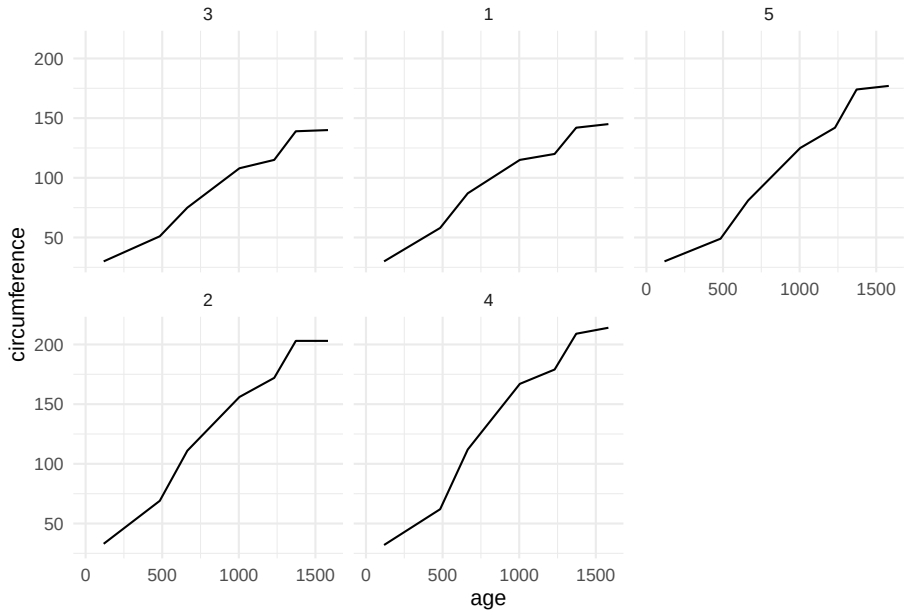
```
glimpse(Orange)
```

```
## Rows: 35
## Columns: 3
## $ Tree          <ord> 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3,~
## $ age           <dbl> 118, 484, 664, 1004, 1231, 1372, 1582, 118, 484, 664, 10~
## $ circumference <dbl> 30, 58, 87, 115, 120, 142, 145, 33, 69, 111, 156, 172, 2~
```

```
ggplot(Orange, aes(x = age, y = circumference, group = Tree,
                   col = Tree)) +
  geom_line()+
  stat_summary(aes(group = 1), fun = mean, col = "red",
               geom = "line", size = 1, show.legend = FALSE,
               linetype = 2)+
  xlim(0, 1600)+
  theme_minimal()
```




```
ggplot(Orange, aes(x = age, y = circumference, group = Tree)) +
  geom_line()+
  xlim(0, 1600)+
  facet_wrap(~Tree)+
  theme_minimal()+
  theme(legend.position = "none")
```



3.11 plotly

plotly cran

Interactive web-based data visualization with R, plotly, and shiny

Plotly R Open Source Graphing Library

```
ggplotly(v1)
ggplotly(v2)
ggplotly(v4)
ggplotly(v5)
```

3.12 esquisse

Alguns links de interesse

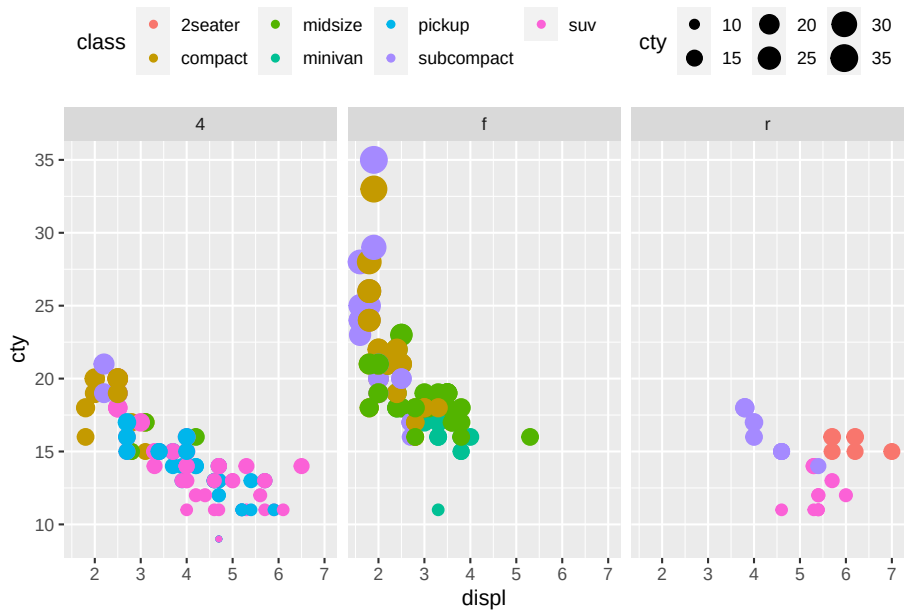
esquisse

esquisse + shiny

```
esquisser(dados)
```

3.13 Exemplo esquisse

```
ggplot(dados) +
  aes(x = displ, y = cty, colour = class, size = cty) +
  geom_point(shape = "circle") +
  scale_color_hue(direction = 1) +
  theme(legend.position = "top") +
  facet_wrap(vars(drv))
```



Capítulo 4

Referências (on-line)

4.1 Livros

ggplot2: elegant graphics for data analysis, Hadley Wickham

R Programming for Data Science, Roger D. Peng

R for Data Science, Hadley Wickham e Garrett Grolemund.

R Graphics Cookbook, Winston Chang

4.2 Tidyverse: links do dplyr, ggplot2 e magrittr

dplyr do tidyverse

ggplot2 do tidyverse

magrittr do tidyverse?

4.3 cheat sheet

Data transformation with dplyr

Data visualization with ggplot2